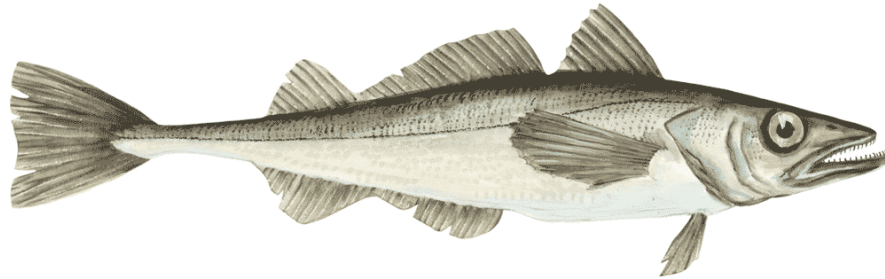


Status of the Pacific Hake (whiting) stock in U.S. and Canadian waters in 2026



Joint Technical Committee of the Pacific Hake/Whiting Agreement
Between the Governments of the United States and Canada

February 19th, 2026

This document reports the collaborative efforts of the official U.S. and Canadian members of the Joint Technical Committee, and others that contributed significantly.

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ONE-PAGE SUMMARY

- The 2026 stock assessment for Pacific Hake uses the model from the 2025 assessment, projected forward to the start of 2026 using the coast-wide fishery catch from 2025. Unlike the usual Pacific Hake stock assessment, the model is not refit to the most recent year (2025) of data.
- The biomass estimate and age data from the 2025 Joint U.S.-Canada Integrated West Coast Pelagics Survey were not available in time to be incorporated into a fully updated assessment model, as a result of the extended U.S. federal government shutdown in 2025. Furthermore, it has yet to be determined how to incorporate the new biomass estimate and data into the existing survey time series in a compatible way, because of changes in survey design.
- The 2025 assessment model, with data through 2024, was previously fit to an acoustic survey index of biomass, annual commercial catch data, and age-composition data from the survey and commercial fisheries. A model-based approach was used for the input weight-at-age matrix and time-varying temperature-dependent maturity matrix, both of which informed fecundity. Fishery catch is the only updated 2025 data used in the 2026 stock assessment for calculations.
- Coast-wide catch in 2025 was 243,298 t (t represents tonnes), 26% below the average over the most recent 10 years (330,442 t), out of a total allowable catch (TAC), adjusted for carryovers, of 400,000 t. The U.S. caught 239,385 t (81.0% of their quota) and Canada caught 3,913 t (3.7% of their quota).
- The median of the relative spawning biomass in 2026 (female spawning biomass at the start of 2026 divided by that at unfished equilibrium, B_0) is 58% but it is highly uncertain (with 95% credible interval from 24% to 138%).
- The median of female spawning biomass at the start of 2026 is 1,066,200 t (with 95% credible interval from 415,262 to 2,728,108 t).
- The probability that female spawning biomass at the start of 2026 is below the $B_{40\%}$ (40% of B_0) reference point is 20.0%, and the probability that relative fishing intensity exceeded 1 in 2025 is 3.0%. The joint probability of both these occurring is 2.9%.
- Based on the default harvest rule, the median catch limit for 2026 is 559,334 t (with 95% credible interval from 182,842 to 1,667,886 t).
- Projections were conducted across a wide range of catch levels for a single year. The projection setting the 2026 catch equal to the 2025 coast-wide TAC of 400,000 t shows the estimated median relative spawning biomass decreasing from 58% in 2026 to 44% in 2027. If a catch of 250,000 t is realized in 2026, it was estimated that the stock has a 33% chance of falling below $B_{40\%}$ at the start of 2027, given the current assumption of median recruitment from 2023 onward.
- Despite depiction of a current stock status likely above $B_{40\%}$, the recent lack of survey abundance and fishery catch in Canada suggests a population structure not conducive to achieving harvest quotas in northern fisheries over recent years.

1 INTRODUCTION

This assessment reports the status of the coastal Pacific Hake (or Pacific whiting, *Merluccius productus*) stock off the west coast of the United States (U.S.) and Canada at the start of 2026. This stock exhibits seasonal migratory behavior, ranging from offshore and generally southern waters during the winter spawning season to coastal areas between northern California and northern British Columbia during the spring, summer, and fall when the fishery is conducted. The stock tends to move farther to the north during the summer in years with warmer water compared to years with colder waters. Older Pacific Hake tend to migrate farther north than younger Pacific Hake in all years, with catches in Canadian waters typically consisting of fish greater than four years old. Separate, and much smaller, populations of Pacific Hake occurring in the major inlets of the Northeast Pacific Ocean, including the Strait of Georgia, Puget Sound, and the Gulf of California, are not included in this analysis.

The 2026 base model structure and data for this Joint Technical Committee (JTC) assessment are the same as the 2025 stock assessment (Johnson et al. 2025), projected forward to the start of 2026 using the fishery catches from 2025. This type of stock assessment is often referred to as a catch-only projection. It is a widespread method used to provide updated management advice given updates to recent catch history while using an already approved stock assessment. Available data at the time of this assessment are presented in Appendix A and Appendix B for reference.

The 2025 base model was a statistical catch-at-age population model fit in a Bayesian context using Markov chain Monte Carlo (MCMC) sampling via the efficient No-U-Turn Sampler (Johnson et al. 2025). The model assumed that the Pacific Hake population is a single coast-wide stock subject to one aggregated fleet with combined male and female population dynamics. The 2026 model was re-fit using MCMC sampling to conduct projections. The stochastic nature of finite MCMC sampling across runs when chains are run in parallel produced similar, but not identical, parameter estimates between the 2025 and 2026 models. However, all parameter estimates are effectively the same as given in Johnson et al. (2025) in terms of population estimates. The 2026 model is again configured using Stock Synthesis (Methot, Jr. and Wetzel 2013) version 3.30.23.1.

A catch-only projection assessment for 2026 was chosen because the biomass estimate and age data from the 2025 Joint U.S.-Canada Integrated West Coast Pelagics Survey were not available in time to be incorporated into a fully updated assessment, as a result of the extended U.S. federal government shutdown in 2025. Furthermore, it has yet to be determined how to incorporate the new biomass estimate and age data into the existing survey time series in a compatible way, because of changes in survey design in 2025. Thus, the JTC decided that conducting the usual fully updated assessment (i.e., adding fishery catch and age data for 2025 and estimating all parameters) without complete survey information would not be appropriate. Such an assessment would be based solely on fishery-age composition data for two consecutive years, whereas usually this only occurs for a single year (in-between the biannual survey). The absence of new survey information to inform the model could considerably increase the risk of inaccurate assessment results being used to inform management decisions. While quantities of interest could still have

been calculated and presented, this could be potentially misleading because numerical estimates of uncertainty would ignore the major structural uncertainty of having no recent survey biomass and age data.

In this document we present catch data, figures and tables related to results from the 2026 base model projections, a Glossary, U.S. and Canadian Advisory Panel reports summarizing the 2025 fishery, and an updated Pacific Hake-specific ecosystem summary. We refer readers to Johnson et al. (2025) for details regarding other data summaries, parameter estimates, model diagnostics, sensitivity analyses, etc. that have effectively not changed from the 2025 assessment.

2 CATCHES

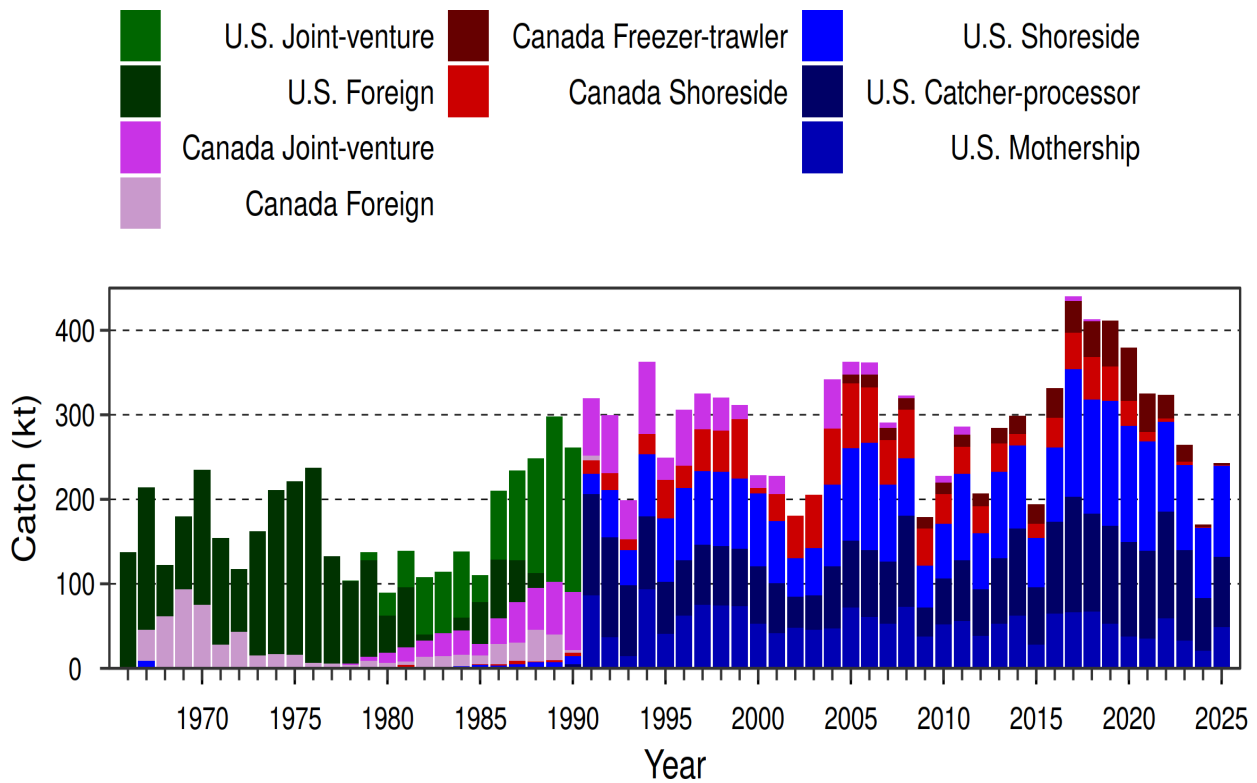


Figure 1. Total Pacific Hake catch used in the assessment by sector, 1966–2025. U.S. tribal catches are included in the sectors where they are represented.

Coast-wide fishery landings of Pacific Hake averaged 242,041 t from 1966 to 2025, with a low of 89,930 t in 1980 and a peak of 440,563 t in 2017 (Figure 1). Prior to 1966, total removals were negligible compared to the modern fishery. Over the early period (1966–1990) most removals were from foreign or joint-venture fisheries. Across the time series, annual catch in U.S. waters averaged 186,563 t (77.1% of the total catch), while catch from Canadian waters averaged 55,478 t. Over the last 10 years, 2016–2025 (Table 1), the average coast-wide catch was 330,442 t with U.S. and Canadian catches averaging 274,439 t and 56,004 t, respectively. Since the peak in 2017, the coast-wide catch declined annually until 2024, with catches increasing in 2025 to 243,298 t out of a total allowable catch (TAC, adjusted

for carryovers) of 400,000 t. Attainment in the U.S. was 81.0% of its quota and in Canada it was 3.7% in 2025.

Table 1. Total Pacific Hake catch used in the assessment by sector for the most recent ten years. U.S. tribal catches are included in the sectors where they are represented.

Year	U.S. Mother-ship	U.S. Catcher-processor	U.S. Shore-side	U.S. Research	U.S. Total	Canada Joint-venture	Canada Shore-side	Canada Freezer-trawler	Canada Total	Total
2016	65,036	108,786	87,762	745	262,328	0	35,012	34,731	69,743	332,071
2017	66,428	136,960	150,454	0	353,842	5,608	43,427	37,686	86,721	440,563
2018	67,121	116,073	134,633	0	317,827	2,724	50,747	41,942	95,413	413,240
2019	52,646	116,146	147,830	0	316,622	0	40,794	54,218	95,013	411,635
2020	37,978	111,147	137,568	95	286,788	0	30,085	62,404	92,489	379,277
2021	35,208	104,030	129,204	917	269,359	0	11,269	45,807	57,076	326,435
2022	59,516	126,247	105,934	0	291,697	0	3,868	27,803	31,671	323,368
2023	33,074	107,117	100,397	0	240,588	0	3,657	20,296	23,952	264,540
2024	21,034	61,914	83,002	0	165,950	0	1,185	2,863	4,048	169,999
2025	48,937	82,625	107,823	0	239,385	0	900	3,013	3,913	243,298

In this document, the terms catch and landings are used interchangeably. Estimates of discard within the target fishery are included but discarding of Pacific Hake in non-target fisheries is not. Discard from all fisheries, including those that do not target Pacific Hake, is estimated to be less than 1% of landings in recent years. The coast-wide catches in 2025 are near the long-term average catch (242,041 t), but, particularly in Canada, the catches have sharply declined over recent years.

3 STOCK BIOMASS AND RECRUITMENT

Results from the base model indicate that since the 1960s, Pacific Hake female spawning biomass has ranged from well below to above unfished equilibrium (Figure 2); see Johnson et al. (2025) for historical details. The median of the 2026 female spawning biomass is 1.066 Mt (with a 95% posterior credibility interval from 0.415 to 2.728 Mt). All 95% credibility intervals are calculated as the 2.5% and 97.5% quantiles of posterior distributions from Markov chain Monte Carlo samples, to give equal-tailed intervals.

The median of the 2026 relative spawning biomass (spawning biomass at the start of 2026 divided by that at unfished equilibrium, B_0) is 58%. However, the uncertainty is large, with a 95% posterior credibility interval from 24% to 138%, partly due to remaining unknowns about the size of the potentially large 2021 cohort because it has still only been observed once by the acoustic survey.

The probability that female spawning biomass at the start of 2026 is below the $B_{40\%}$ (40% of B_0) reference point is 20.0%, and the probability that relative fishing intensity exceeded 1 in 2025 is 3.0%. The joint probability of both these occurring is 2.9%.

Pacific Hake have low to moderate recruitment (relative to the long-term average) with occasional large year classes (Figure 3). Estimates of recruitment are similar to those from Johnson et al. (2025). The size of the 2021 cohort remains highly uncertain due to limited data (e.g., compare the uncertainty to that of the 2016 cohort, a cohort that has a comparable median).

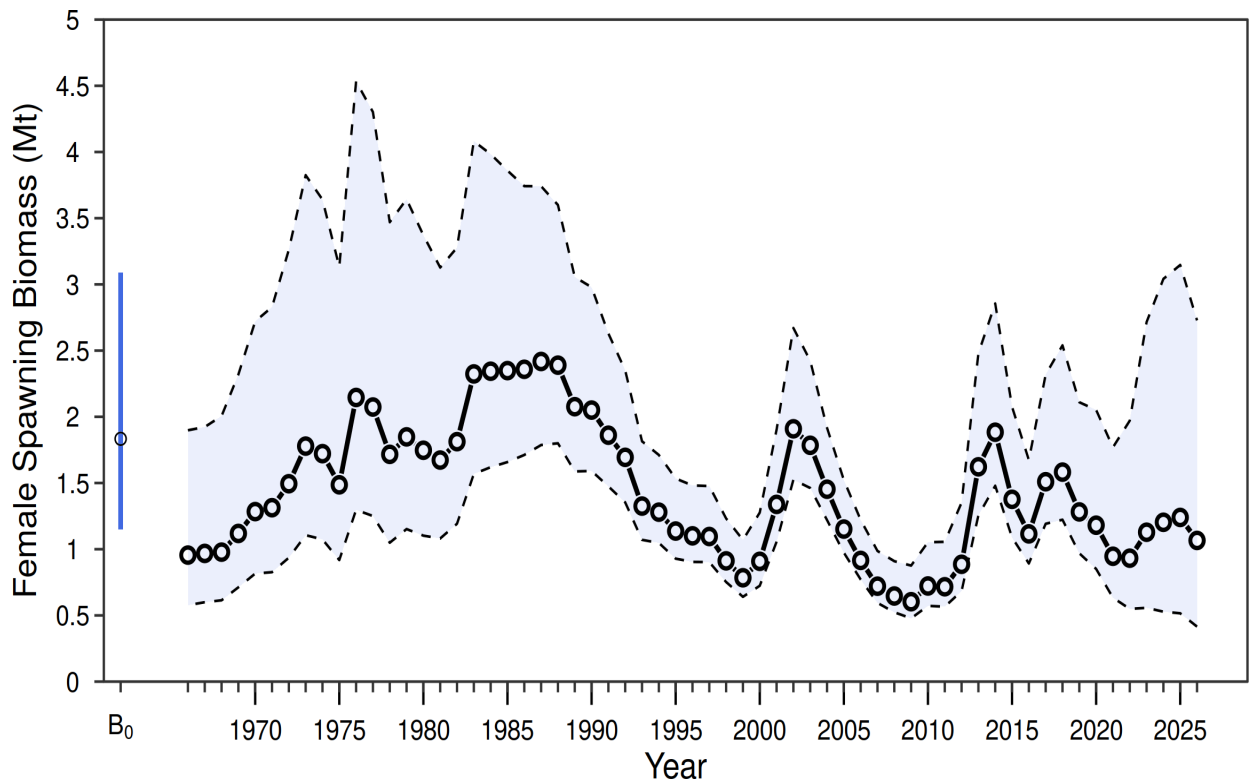


Figure 2. Median (solid line) of the posterior distribution for beginning of the year female spawning biomass (B_t in year t ; Mt) through 2026 with 95% posterior credibility intervals (shaded area). The left-most circle with a 95% posterior credibility interval is the estimated unfished equilibrium biomass, B_0 .

4 MANAGEMENT PERFORMANCE

Over the last decade (2016–2025), the mean coast-wide utilization rate (proportion of catch target removed) has been 61.2% and catches have been below coast-wide targets (Table 2). From 2021 to 2025, the mean utilization rates differed between the U.S. (63.8%) and Canada (19.6%). While Canada’s rate was higher than the U.S.’s in 2020, it has dropped considerably over the past several years to historic lows. The utilization rate for the coast-wide fishery this year (60.8%) increased relative to the last two years which had low attainment due to difficulties locating aggregations of fish during the spring and throughout the year in northern areas. The usual 73.88% and 26.12% allocation of coast-wide TAC, as specified in the Joint U.S.-Canada Agreement for Pacific Hake, was implemented for the 2025 fishery. The U.S. proportion of the coast-wide catch has exceeded 90% in each of the last four years. Total landings last exceeded the coast-wide quota in 2002 when utilization was 112%.

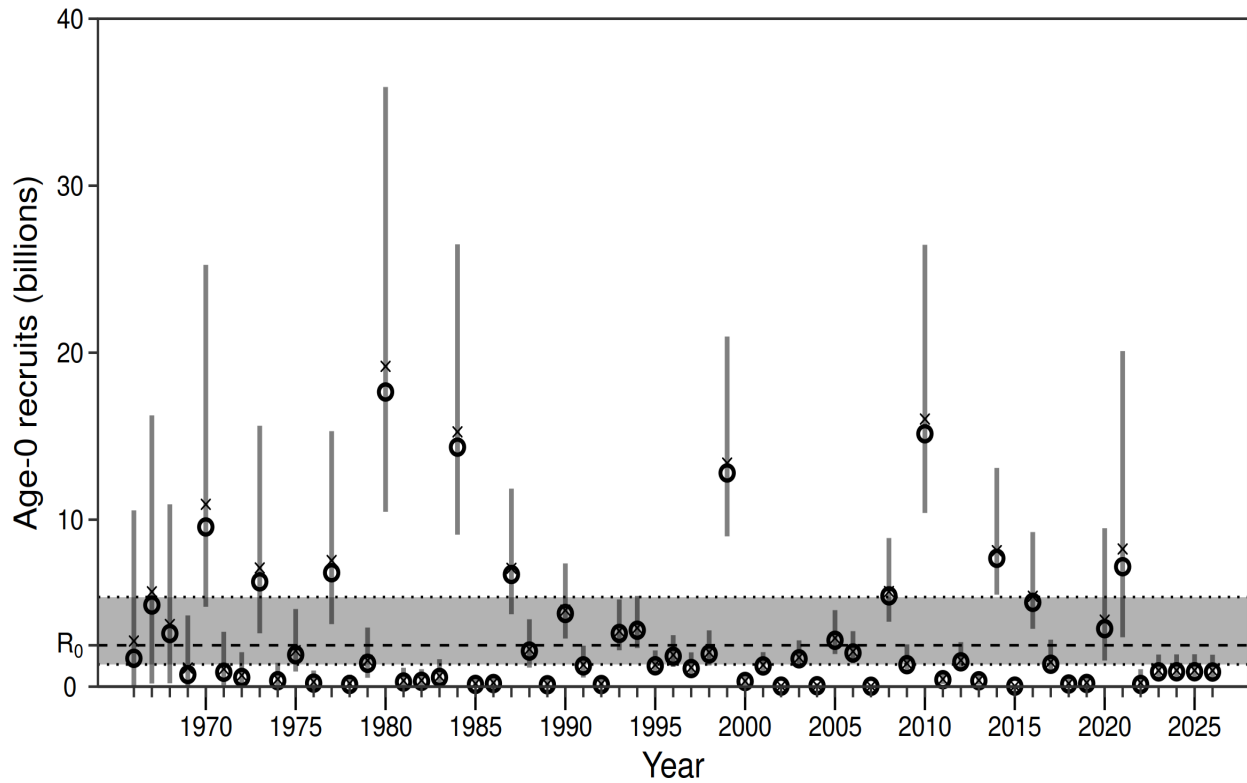


Figure 3. Medians (solid circles) and means (X) of the posterior distribution for recruitment (billions of age-0 fish) with 95% posterior credibility intervals (vertical lines). The median of the posterior distribution for mean unfished equilibrium recruitment (R_0) is shown as the horizontal dashed line with the 95% posterior credibility interval shaded between the dotted lines.

Table 2. Recent trends in Pacific Hake landings and management decisions. Catch targets in 2020 and 2021 were specified unilaterally. All landings and catch targets are given in tonnes.

Year	U.S. landings	Canada landings	Total landings	U.S. prop. of total catch	Canada prop. of total catch	U.S. catch target	Canada catch target	Total catch target	U.S. prop. of catch target removed	Canada prop. of catch target removed	Total prop. of catch target removed
2016	262,328	69,743	332,071	79.0%	21.0%	367,553	129,947	497,500	71.4%	53.7%	66.7%
2017	353,842	86,721	440,563	80.3%	19.7%	441,433	156,067	597,500	80.2%	55.6%	73.7%
2018	317,827	95,413	413,240	76.9%	23.1%	441,433	156,067	597,500	72.0%	61.1%	69.2%
2019	316,622	95,013	411,635	76.9%	23.1%	441,433	156,067	597,500	71.7%	60.9%	68.9%
2020	286,788	92,489	379,277	75.6%	24.4%	424,810	104,480	529,290	67.5%	88.5%	71.7%
2021	269,359	57,076	326,435	82.5%	17.5%	369,400	104,480	473,880	72.9%	54.6%	68.9%
2022	291,697	31,671	323,368	90.2%	9.8%	402,646	142,354	545,000	72.4%	22.2%	59.3%
2023	240,588	23,952	264,540	90.9%	9.1%	461,750	163,250	625,000	52.1%	14.7%	42.3%
2024	165,950	4,048	169,999	97.6%	2.4%	410,034	144,966	555,000	40.5%	2.8%	30.6%
2025	239,385	3,913	243,298	98.4%	1.6%	295,520	104,480	400,000	81.0%	3.7%	60.8%

5 FORECAST DECISION TABLES

For the 2025 stock assessment (Johnson et al. 2025), the spawning biomass was estimated at the start of 2025, and then projections were done assuming different catch scenarios that specified catches in 2025 (now known) and 2026 (assumed alternatives). This gave projected biomass quantities at the start of 2026. The recruitment deviations were set to 0 for 2023 to 2026. This covers the two years (2023 and 2024) for which no survey age data

was used to fit the model, and the two projection years (2025 and 2026). This was a change from previous stock assessments, in which deviations were drawn from a distribution that could allow very large (or very small) recruitments to occur and propagate through the projection years as the cohorts matured. The change to no recruitment deviations was deemed preferable for the 2025 assessment for several reasons, including that the “information on the size of the 2023 year class from the 2024 fishery was considered suspect given the unique fishing conditions during the year” (Johnson et al. 2025). Allowing for recruitment deviations during forecast years could have allowed “potentially large recruitments that can inflate the forecast biomass yet are based on limited or no data and so may not materialise in practice” (Johnson et al. 2025). We continue to apply no recruitment deviations for the 2026 base model for 2023 onwards as we are using the same assumptions as for the 2025 base model, and we are not fitting to new data.

The assumption of not allowing for recruitment deviations in forecasts did cause the projected populations to decline for the 2025 assessment (Johnson et al. 2025) even with 0 t catch in the projection years. This is because the assumption removes the lognormal (skewed) distribution of recruitments, with occasional large recruitments that can pull the stock out of a decline, that has been seen historically (Figure 3). Instead, average recruitment (i.e., the median resulting from the bias-corrected mean of the lognormal distribution) leads to a constant recruitment level close to $0.38R_0$ (Figure 3), and does not allow for the occasional large recruitments seen when applying the lognormal distribution of recruitment deviations (e.g., see the broader distribution of potential 2023 and 2024 recruitments shown in Figure f of the 2024 assessment, Grandin et al. 2024).

For this assessment, we present projections for just one year, rather than three years as for previous assessments. Projections assume various catch scenarios for 2026 and then forecast the biomass at the start of 2027. This was done for two reasons:

1. As just outlined, the assumption of not using recruitment deviations during the forecast period is not a feasible long-term model assumption because the median recruitment is too low to sustain the population in perpetuity. Forecasting further years ahead will exacerbate this and lead to unrealistic results given past levels of Pacific Hake recruitment.
2. The models in the 2024 and 2025 assessments both estimated a stock biomass somewhat larger than the biomass index from the 2023 survey; Figure 18 of Grandin et al. (2024) and Figure 17 of Johnson et al. (2025). Inclusion of the 2025 survey biomass index would have helped to determine the validity of this overfitting. Without the 2025 survey biomass index, it was deemed pragmatic to not overextend the forecasts too far ahead from 2023 survey.

In future assessments, the JTC will again consider reimplementing recruitment deviations and return to providing multiple forecast years, as the model should be fit to more up-to-date data (including the 2025 survey biomass and age data).

The catch limit for 2026 based on the default $F_{40\%}-40:10$ harvest policy has a median of 559,334 t with a wide range of uncertainty, the 95% credibility interval being 182,842–1,667,886 t.

Table 3. Forecast quantiles of Pacific Hake relative spawning biomass at the beginning of the year. Catch alternatives are defined by letters a-j. TAC: total allowable catch. HR: harvest rate.

Catch alternative	Catch year	Catch (t)	Biomass at start of year	Relative spawning biomass		
				5%	50%	95%
			Start of 2026	0.28	0.58	1.18
a:	2026	0	Start of 2027	0.29	0.55	1.05
b:	2026	150,000	Start of 2027	0.25	0.51	1.01
c:	2026	200,000	Start of 2027	0.24	0.49	1.00
d:	2026	250,000	Start of 2027	0.23	0.48	0.98
e:	2026	300,000	Start of 2027	0.21	0.47	0.97
f:	2026	350,000	Start of 2027	0.20	0.45	0.95
g (2025 TAC):	2026	400,000	Start of 2027	0.19	0.44	0.94
h:	2026	450,000	Start of 2027	0.17	0.43	0.93
i:	2026	500,000	Start of 2027	0.16	0.42	0.91
j (Default HR):	2026	559,334	Start of 2027	0.15	0.40	0.90

Table 4. Forecast quantiles of Pacific Hake relative fishing intensity $(1 - \text{SPR}) / (1 - \text{SPR}_{40\%})$, expressed as a proportion. Catch alternatives are defined by letters a-j. TAC: total allowable catch. HR: harvest rate.

Catch alternative	Catch year	Catch (t)	Relative fishing intensity		
			5%	50%	95%
a:	2026	0	0.00	0.00	0.00
b:	2026	150,000	0.23	0.47	0.79
c:	2026	200,000	0.30	0.56	0.91
d:	2026	250,000	0.35	0.64	1.00
e:	2026	300,000	0.40	0.71	1.07
f:	2026	350,000	0.45	0.77	1.13
g (2025 TAC):	2026	400,000	0.49	0.82	1.18
h:	2026	450,000	0.53	0.87	1.23
i:	2026	500,000	0.57	0.91	1.26
j (Default HR):	2026	559,334	0.61	0.96	1.29

Decision tables give the projected population status (i.e., relative spawning biomass) and fishing intensity relative to the target under different catch alternatives for the base model (Tables 3 and 4). The tables are organized to show the projected outcome for each potential catch level and year (row) across the quantiles (columns) of the posterior distribution. Tables show results for one year of future catch levels based on subsequent projected stock status and fishing intensity. Figure 4 shows the projected relative spawning biomass for several of the catch alternatives. Population dynamics and governing parameters assumed during the forecast period include selectivity, weight-at-age and fecundity-at-age averaged over the five most recent years of data (2020–2024), and constant values for all other parameters; see Johnson et al. (2025).

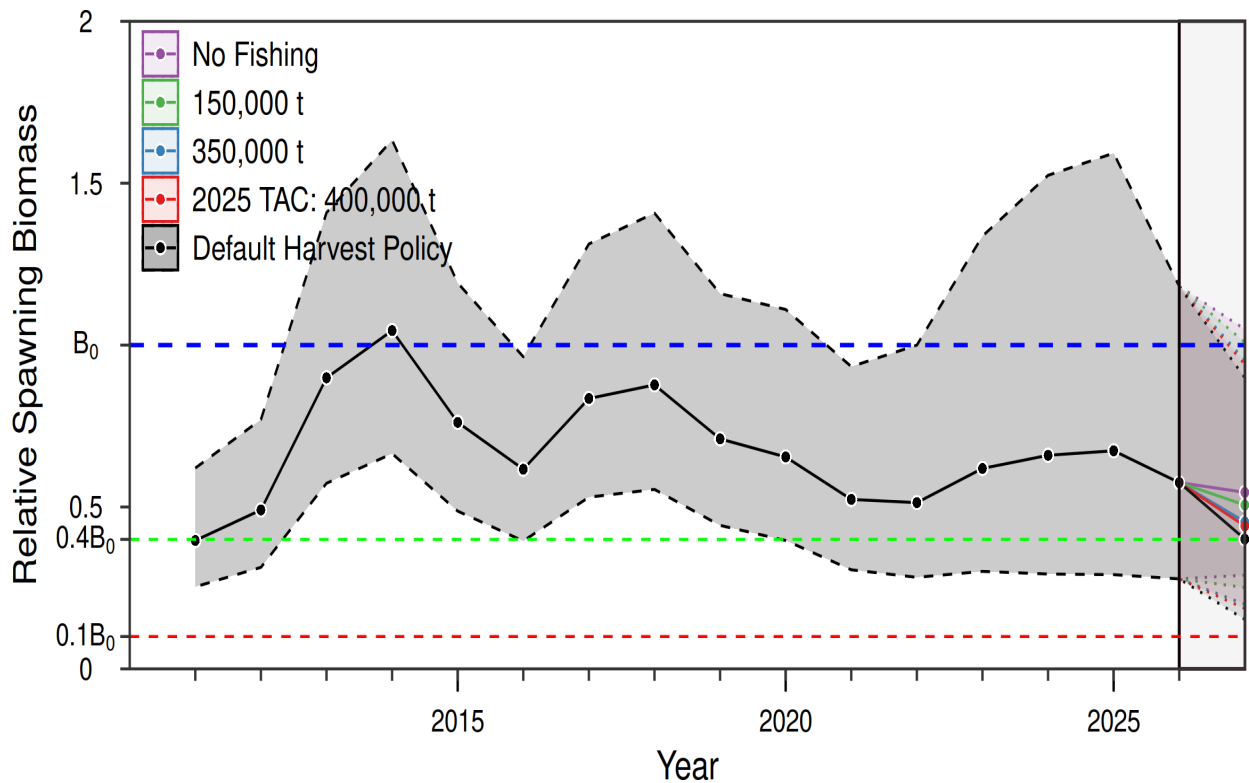


Figure 4. Median and 95% posterior credibility intervals of relative spawning biomass to the start of 2026 from the base model and projections to the start of 2027 for several catch alternatives, which are defined in Table 3.

A relative fishing intensity of 1 should indicate fishing at the $F_{\text{SPR}=40\%}$ default harvest rate catch target. But the projected median relative fishing intensity can be slightly different than the target because the $F_{\text{SPR}=40\%}$ default harvest-rate catch limit is calculated using baseline selectivity-at-age (1966–1990; prior to time-varying deviations), whereas the forecasted catches use selectivity averaged over the last five years. Recent changes in selectivity will thus be reflected in the determination of fishing relative to the default harvest policy. For example, fishing at the $F_{\text{SPR}=40\%}$ default harvest-rate catch limit (scenario j: default HR) in 2026 results in a median relative fishing intensity of 0.96 (Table 4).

With zero catch for 2026 and no recruitment deviations from 2023 onward, the biomass has an 85% probability of decreasing from 2026 to 2027 (Table 5 and Figure 5).

For all explored catches, the maximum probability of female spawning biomass at the start of 2027 dropping below $B_{10\%}$ is 0.6% and of dropping below $B_{40\%}$ is 49.8% (Table 5 and Figure 5). As the above average 2014 and 2016 cohorts continue to age, total biomass of these cohorts even without fishing mortality is expected to decrease as losses from mortality outweigh increases from growth. The estimated above-average 2021 cohort will continue to play a large role in determining female spawning biomass during the forecast year presented here.

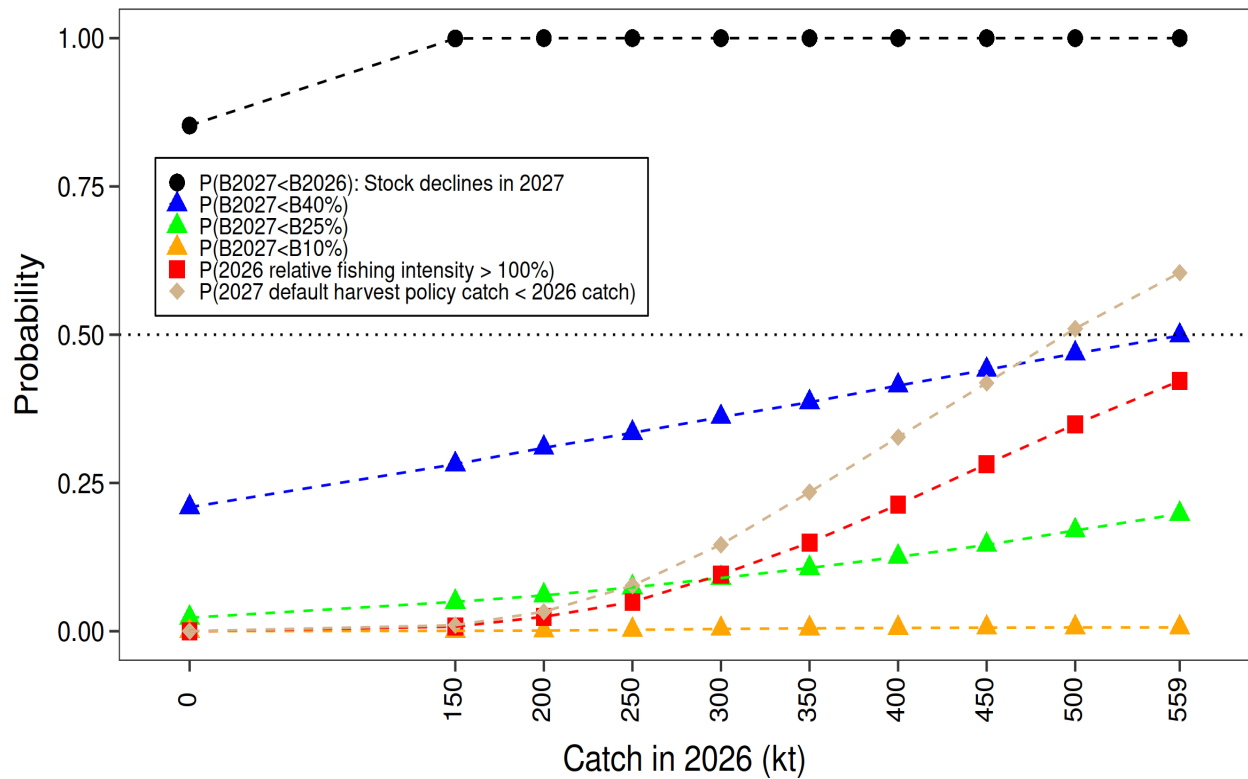


Figure 5. Graphical representation of the probabilities related to spawning biomass, relative fishing intensity, and the 2027 default harvest policy catch for alternative 2026 catch options (explained in Table 3) as listed in Table 5. The symbols indicate points that were computed directly from model output and lines interpolate between the points.

Table 5. Probabilities related to spawning biomass, relative fishing intensity, and the 2027 default harvest policy catch for alternative 2026 catch options (catch options are explained in Table 3).

	Catch (t) in 2026	$B_{2027} < B_{2026}$	$B_{2027} < B_{40\%}$	$B_{2027} < B_{25\%}$	$B_{2027} < B_{10\%}$	2026 Fishing intensity > 100%	2027 Default HR catch < 2026 catch
a:	0	0.85	0.21	0.02	0.00	0.00	0.00
b:	150,000	1.00	0.28	0.05	0.00	0.01	0.01
c:	200,000	1.00	0.31	0.06	0.00	0.02	0.03
d:	250,000	1.00	0.33	0.07	0.00	0.05	0.08
e:	300,000	1.00	0.36	0.09	0.00	0.10	0.15
f:	350,000	1.00	0.39	0.11	0.00	0.15	0.23
g:	400,000	1.00	0.41	0.13	0.01	0.21	0.33
h:	450,000	1.00	0.44	0.15	0.01	0.28	0.42
i:	500,000	1.00	0.47	0.17	0.01	0.35	0.51
j:	559,334	1.00	0.50	0.20	0.01	0.42	0.60

6 RESPONSE TO 2025 SCIENTIFIC REVIEW GROUP (SRG) REVIEW

The Scientific Review Group (SRG) meeting was held from February 10–14th, 2025, as a virtual meeting.

While this year is a projection based assessment, the JTC provides the following responses to the 2025 ‘SRG Recommendations and Conclusions for the Stock Assessment’ to update the SRG about ongoing and future research plans:

1. **The SRG recommends a thorough analysis of catch and biomass distribution for Canada and US that examines latitudinal shifts over time, and seeks to explain poor fishery performance in recent years. The SRG recommends developing a limited set of standardized descriptive spatial distribution metrics (such as center of gravity) to support future year discussions.**

Response – The JTC continues to support an analysis across available data sources to better understand the current distribution of the stock along the U.S. West Coast. The planned work of building species distribution models incorporating environmental factors across the transboundary range of Pacific Hake has been delayed due to staffing changes at the NWFSC.

The JTC is also continuing work evaluating patterns of Pacific Hake in the U.S. West Coast Bottom Trawl Survey in California waters where fishery data are limited. The JTC will also continue work investigating spatial fishing effort, catch, and revenue over time using data summary tools (e.g., [Pacific Fishing Effort Mapping Project](#)).

2. **The SRG recommends continuing sensitivities for steepness, natural mortality, σ_{R^*} , including the age-1 index, alternative standard deviations for time-varying selectivity, and down-weighting fishery age-composition data.**

Response – The recommended set of sensitivity model runs were not applicable for the 2026 base model, which is based solely on a projection of the 2025 base model where the realized total catch in 2025 is specifically used. The standard set of sensitivity model runs was presented for the 2025 base model in the 2025 assessment ([Johnson et al. 2025](#)) and will be incorporated into next year’s stock assessment.

3. **The SRG recommends that the JTC continue exploring alternative ways of estimating natural mortality to update the current approach in the model, and particularly consider options which have age-based natural mortality, and may be external to SS3.**

Response – Pacific Hake have several natural predators, including themselves, particularly on younger ages. Recently, [Trzcinski et al. \(2024\)](#) identified Pacific Hake as the largest mean proportion of the diet of harbor seals. Additionally, the abundance of harbor seals is near carrying capacity in Canadian waters and has been since the late 1990s. Stock Synthesis allows for the inclusion of a predator index to estimate predation mortality in addition to baseline natural mortality. Data for predation mortality can be absolute numbers of dead fish due to the predator, an index of predator effort, and predation composition (age or length) data. A predation index based on estimated abundance of harbor seals could be added to the assessment model, but this has not been explored.

A potential, better path forward is to use Dynamic Structural Equation Modeling (DSEM) to include indices for multiple predators and allow for two-way interactions between them and M . This capability is not, and will not be, available within Stock Synthesis but is possible using models written in Template Model Builder. A proof of concept has been implemented at the Alaska Fisheries Science Center and work is ongoing to implement this in the Fisheries Integrated Modeling System (FIMS).

In 2025, the JTC presented sensitivities to using estimates of age-specific M from the Climate-Enhanced, Age-based model with Temperature-specific Trophic Linkages and Energetics (CEATTLE) model as a way forward (Holsman et al. 2016). Output from CEATTLE includes time series of estimates of baseline M and estimates of M at age for ages 1, 2, 3, 4, and 5, where the M at-age estimates are strictly natural mortality due to cannibalism.

The JTC plans to continue to evaluate how to account for age-based and potentially time-varying mortality in future assessments.

- 4. The SRG recommends examining alternative structural assumptions for the stock assessment. More complex structural assumptions may utilize the data more thoroughly, explain different trends across areas and/or fleets, and estimate stock status more accurately, but simpler models may be more appropriate for determination of the TAC. The MSE can be used to determine best performing assessment models for management. Priority avenues of examinations include multiple fleets (e.g., fleets-as-areas) and alternative parametric and time-varying (or time-invariant) selectivity patterns.**

Response – The JTC plans to continue work to evaluate the multitude of changes in the assessment process itself (e.g., data availability, assumptions, etc.) that would be needed to develop alternative model structures. This includes the development of a research plan, including complimentary or standalone analyses, alternative model structures to explore, and simulation analyses to evaluate and compare alternative models (e.g., using one or more tools, such as MSE).

The JTC has begun the preliminary work of splitting catch and age data starting in 1991 for each country to evaluate an alternative model structure that splits data by country from 1991 onward with all historical data (1966–1990) in a combined U.S./Canada fleet. The JTC will brief the SRG about preliminary work at the 2026 meeting.

Other steps the JTC plans to take in the future include evaluating alternative data and model assumptions that arise for validity given proposed new model structures; evaluating variance and bias trade-offs associated with new parameterizations (e.g., doubling the number of time-varying selectivity parameters when modelling recent fishery data by country); comparing alternative spatially-implicit assessment approaches (e.g., time-varying selectivity versus area-varying selectivity, or both); and developing states of nature scenarios to test model structure robustness in an MSE framework. Additionally, model complexity in the spatial domain needs to be addressed relative to other structural assumptions in the assessment model. Other assumptions may be more influential or have confounding or complimentary impacts on model sensitivity and robustness. These may include data weighting, improving estimates of natural mortality via predator time series, multiple fleets instead of a single aggregated fleet, and using fleet-specific weight-at-age data

instead of aggregated weight-at-age (the JTC has looked at this before, but it hasn't been revisited in over a decade).

Changes to the structure of the assessment model may not be the most immediate need for understanding changes in hake distribution. Some of the other (higher priority) SRG requests noted above will help with understanding fundamental mechanisms, which can help to formulate hypotheses to inform relevant model structures. These include spatial CPUE modeling, investigating hake distribution changes in bottom trawl surveys, and correlating changes in hake distribution with the California Current Integrated Ecosystem Assessment.

- 5. The SRG recommends considering including the eDNA index of abundance for inclusion in the assessment as a second index of abundance for Pacific Hake, also estimating additional variance for the index given the small nominal CV values.**

Response – The JTC will continue to consider the eDNA index in future full assessments. Water samples were collected during the 2025 survey for enumeration of eDNA, providing a fourth year in the eDNA index time series once laboratory processing and subsequent statistical analyses have been completed (expected by Spring 2026).

- 6. The SRG recommends that the JTC continue investigating how to include the age-1 index in the stock assessment in this or future assessment frameworks, but notes that the selectivity/catchability of the gear is expected to change depending on the AWT or MFT gear used in the survey.**

Response – The JTC is following the progress of the net-selectivity work being conducted by the IWCPs team, which will be a critical component to better understand changes in the sampling of age-1 hake and the potential options for including any age-1 index in future stock assessments.

- 7. The SRG encourages the JTC to consider alternative methods to determine the maximum input sample size for the age compositions (e.g. Stewart and Hamel 2014, Hulson et al. 2023).**

Response – The JTC agrees that determining input sample sizes is important for fishery and survey age compositions, which then provides the basis from which model-based re-weighting is done. The JTC has considered alternative methods and has determined additional research is needed to inform the calculation of input sample size for both fishery and survey ages. Fishery input sample sizes are calculated using a mixture of either the number of hauls or trips, when haul information is not available (e.g., shoreside), and further work is needed to determine the effective sample sizes at the haul and trip level and how to calculate a fleet-wide input sample size. Additionally, there is intersection between this research and alternative model structures regarding the number of fleets and the impact of alternative ways to determine input sample size within each fleet which the JTC will consider in ongoing research.

Sensitivity analyses using alternative input sample sizes, following the Stewart and Hamel (2014) approach, will continue to be evaluated in subsequent assessments.

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Having all of the code under git version control has greatly increased the collaboration between authors. It has been in use since the 2016 assessment ([Grandin et al. 2016](#)) and has ensured delivery of the assessment document each year within a contracted time frame.

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A DATA TABLES

This appendix contains empirical data tables only (though also see Appendix G). There are no model output tables included. The 2026 assessment is a simple projection model which includes only the total 2025 catch as new data. The tables presented here contain historical data as well as the data for 2025, even though only catch was updated in the modelling for this year. All the 2025 data presented here will be used in future assessment years' model runs.

Table A.1. Annual catches of Pacific Hake (t) in U.S. waters by fleet, 1966–2025. Tribal catches are included in the fleet totals. Research catch includes landed catch associated with research-related activities. Catch associated with surveys and discarded bycatch in fisheries not targeting hake is not currently included in the table or model.

Year	Foreign	Joint-venture	Mother-ship	Catcher-processor	Shore-side	Research	Total
1966	137,000	0	0	0	0	0	137,000
1967	168,700	0	0	0	8,960	0	177,660
1968	60,660	0	0	0	160	0	60,820
1969	86,190	0	0	0	90	0	86,280
1970	159,510	0	0	0	70	0	159,580
1971	126,490	0	0	0	1,430	0	127,920
1972	74,090	0	0	0	40	0	74,130
1973	147,440	0	0	0	70	0	147,510
1974	194,110	0	0	0	0	0	194,110
1975	205,650	0	0	0	0	0	205,650
1976	231,330	0	0	0	220	0	231,550
1977	127,010	0	0	0	490	0	127,500
1978	96,827	860	0	0	690	0	98,377
1979	114,910	8,830	0	0	940	0	124,680
1980	44,023	27,537	0	0	790	0	72,350
1981	70,365	43,557	0	0	838	0	114,760
1982	7,089	67,465	0	0	1,023	0	75,577
1983	0	72,100	0	0	1,051	0	73,151
1984	14,772	78,889	0	0	2,721	0	96,382
1985	49,853	31,692	0	0	3,894	0	85,439
1986	69,861	81,640	0	0	3,432	0	154,932
1987	49,656	105,997	0	0	4,795	0	160,448
1988	18,041	135,781	0	0	6,867	0	160,690
1989	0	195,636	0	0	7,414	0	203,049
1990	0	170,972	0	4,537	9,632	0	185,142
1991	0	0	86,408	119,411	23,970	0	229,789
1992	0	0	36,721	117,981	56,127	0	210,829
1993	0	0	14,558	83,466	42,108	0	140,132
1994	0	0	93,610	86,251	73,616	0	253,477
1995	0	0	40,805	61,357	74,962	0	177,124
1996	0	0	62,098	65,933	85,128	0	213,159
1997	0	0	75,128	70,832	87,416	0	233,376
1998	0	0	74,686	70,377	87,856	0	232,920
1999	0	0	73,440	67,655	83,470	0	224,565
2000	0	0	53,110	67,805	85,854	0	206,770
2001	0	0	41,901	58,628	73,412	0	173,940
2002	0	0	48,404	36,342	45,708	0	130,453
2003	0	0	45,396	41,214	55,335	0	141,945
2004	0	0	47,561	73,176	96,503	0	217,240
2005	0	0	72,178	78,890	109,052	0	260,120
2006	0	0	60,926	78,864	127,165	0	266,955
2007	0	0	52,977	73,263	91,441	0	217,682
2008	0	0	72,440	108,195	67,760	0	248,395
2009	0	0	37,550	34,800	49,222	0	121,573

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Year	Foreign	Joint-venture	Mother-ship	Catcher-processor	Shore-side	Research	Total
2010	0	0	52,022	54,291	64,653	0	170,967
2011	0	0	56,394	71,678	102,146	1,042	231,261
2012	0	0	38,512	55,264	65,920	448	160,145
2013	0	0	52,470	77,950	102,141	1,018	233,578
2014	0	0	62,102	103,203	98,640	197	264,141
2015	0	0	27,665	68,484	58,011	0	154,160
2016	0	0	65,036	108,786	87,762	745	262,328
2017	0	0	66,428	136,960	150,454	0	353,842
2018	0	0	67,121	116,073	134,633	0	317,827
2019	0	0	52,646	116,146	147,830	0	316,622
2020	0	0	37,978	111,147	137,568	95	286,788
2021	0	0	35,208	104,030	129,204	917	269,359
2022	0	0	59,516	126,247	105,934	0	291,697
2023	0	0	33,074	107,117	100,397	0	240,588
2024	0	0	21,034	61,914	83,002	0	165,950
2025	0	0	48,937	82,625	107,823	0	239,385

Table A.2. Annual catches of Pacific Hake (t) in Canadian waters by fleet, 1966–2025.

Year	Foreign	Joint-venture	Shore-side	Freezer-trawler	Total
1966	700	0	0	0	700
1967	36,710	0	0	0	36,710
1968	61,360	0	0	0	61,360
1969	93,850	0	0	0	93,850
1970	75,010	0	0	0	75,010
1971	26,700	0	0	0	26,700
1972	43,410	0	0	0	43,410
1973	15,130	0	0	0	15,130
1974	17,150	0	0	0	17,150
1975	15,700	0	0	0	15,700
1976	5,970	0	0	0	5,970
1977	5,190	0	0	0	5,190
1978	3,450	1,810	0	0	5,260
1979	7,900	4,230	300	0	12,430
1980	5,270	12,210	100	0	17,580
1981	3,920	17,160	3,280	0	24,360
1982	12,480	19,680	0	0	32,160
1983	13,120	27,660	0	0	40,780
1984	13,200	28,910	0	0	42,110
1985	10,530	13,240	1,190	0	24,960
1986	23,740	30,140	1,770	0	55,650
1987	21,450	48,080	4,170	0	73,700
1988	38,080	49,240	830	0	88,150
1989	29,750	62,718	2,562	0	95,029
1990	3,810	68,314	4,021	0	76,144
1991	5,610	68,133	16,174	0	89,917
1992	0	68,779	20,043	0	88,822
1993	0	46,422	12,352	0	58,773
1994	0	85,154	23,776	0	108,930
1995	0	26,191	46,181	0	72,372
1996	0	66,779	26,360	0	93,139

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Year	Foreign	Joint-venture	Shore-side	Freezer-trawler	Total
1997	0	42,544	49,227	0	91,771
1998	0	39,728	48,074	0	87,802
1999	0	17,201	70,121	0	87,322
2000	0	15,625	6,382	0	22,007
2001	0	21,650	31,935	0	53,585
2002	0	0	50,244	0	50,244
2003	0	0	63,217	0	63,217
2004	0	58,892	66,175	0	125,067
2005	0	15,695	77,335	9,985	103,014
2006	0	14,319	65,289	15,136	94,744
2007	0	6,780	52,649	14,121	73,550
2008	0	3,592	57,795	13,214	74,602
2009	0	0	44,130	13,223	57,353
2010	0	8,081	35,362	13,573	57,016
2011	0	9,717	31,760	14,596	56,073
2012	0	0	32,147	14,912	47,059
2013	0	0	33,665	18,584	52,249
2014	0	0	13,326	21,792	35,118
2015	0	0	16,775	22,909	39,684
2016	0	0	35,012	34,731	69,743
2017	0	5,608	43,427	37,686	86,721
2018	0	2,724	50,747	41,942	95,413
2019	0	0	40,794	54,218	95,013
2020	0	0	30,085	62,404	92,489
2021	0	0	11,269	45,807	57,076
2022	0	0	3,868	27,803	31,671
2023	0	0	3,657	20,296	23,952
2024	0	0	1,185	2,863	4,048
2025	0	0	900	3,013	3,913

Table A.3. Pacific Hake landings and management decisions, 1966–2025. A dash (–) indicates the management decision was either not specified or was unknown to the authors at the time of this assessment.

Year	U.S. landings	Canada landings	Total landings	U.S. prop. of total catch	Canada prop. of total catch	U.S. catch target	Canada catch target	Total catch target	U.S. prop. of catch target removed	Canada prop. of catch target removed	Total prop. of catch target removed
1966	137,000	700	137,700	99.5%	0.5%	–	–	–	–	–	–
1967	177,660	36,710	214,370	82.9%	17.1%	–	–	–	–	–	–
1968	60,820	61,360	122,180	49.8%	50.2%	–	–	–	–	–	–
1969	86,280	93,850	180,130	47.9%	52.1%	–	–	–	–	–	–
1970	159,580	75,010	234,590	68.0%	32.0%	–	–	–	–	–	–
1971	127,920	26,700	154,620	82.7%	17.3%	–	–	–	–	–	–
1972	74,130	43,410	117,540	63.1%	36.9%	–	–	–	–	–	–
1973	147,510	15,130	162,640	90.7%	9.3%	–	–	–	–	–	–
1974	194,110	17,150	211,260	91.9%	8.1%	–	–	–	–	–	–
1975	205,650	15,700	221,350	92.9%	7.1%	–	–	–	–	–	–
1976	231,550	5,970	237,520	97.5%	2.5%	–	–	–	–	–	–
1977	127,500	5,190	132,690	96.1%	3.9%	–	–	–	–	–	–
1978	98,377	5,260	103,637	94.9%	5.1%	130,000	–	–	75.7%	–	–
1979	124,680	12,430	137,110	90.9%	9.1%	198,900	35,000	–	62.7%	35.5%	–
1980	72,350	17,580	89,930	80.5%	19.5%	175,000	35,000	–	41.3%	50.2%	–
1981	114,760	24,360	139,120	82.5%	17.5%	175,000	35,000	–	65.6%	69.6%	–
1982	75,577	32,160	107,737	70.1%	29.9%	175,000	35,000	–	43.2%	91.9%	–
1983	73,151	40,780	113,931	64.2%	35.8%	175,000	45,000	–	41.8%	90.6%	–
1984	96,382	42,110	138,492	69.6%	30.4%	175,000	45,000	270,000	55.1%	93.6%	51.3%
1985	85,439	24,960	110,399	77.4%	22.6%	175,000	50,000	212,000	48.8%	49.9%	52.1%
1986	154,932	55,650	210,582	73.6%	26.4%	295,800	75,000	405,000	52.4%	74.2%	52.0%
1987	160,448	73,700	234,148	68.5%	31.5%	195,000	75,000	264,000	82.3%	98.3%	88.7%
1988	160,690	88,150	248,840	64.6%	35.4%	232,000	98,000	327,000	69.3%	89.9%	76.1%
1989	203,049	95,029	298,079	68.1%	31.9%	225,000	98,000	323,000	90.2%	97.0%	92.3%
1990	185,142	76,144	261,286	70.9%	29.1%	196,000	73,500	245,000	94.5%	103.6%	106.6%
1991	229,789	89,917	319,705	71.9%	28.1%	228,000	98,000	253,000	100.8%	91.8%	126.4%
1992	210,829	88,822	299,650	70.4%	29.6%	208,800	90,000	232,000	101.0%	98.7%	129.2%
1993	140,132	58,773	198,905	70.5%	29.5%	142,000	61,000	178,000	98.7%	96.3%	111.7%
1994	253,477	108,930	362,407	69.9%	30.1%	260,000	110,000	325,000	97.5%	99.0%	111.5%
1995	177,124	72,372	249,495	71.0%	29.0%	178,400	76,500	223,000	99.3%	94.6%	111.9%
1996	213,159	93,139	306,299	69.6%	30.4%	212,000	91,000	265,000	100.5%	102.4%	115.6%

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Year	U.S. landings	Canada landings	Total landings	U.S. prop. of total catch	Canada prop. of total catch	U.S. catch target	Canada catch target	Total catch target	U.S. prop. of catch target removed	Canada prop. of catch target removed	Total prop. of catch target removed
1997	233,376	91,771	325,147	71.8%	28.2%	232,000	99,400	290,000	100.6%	92.3%	112.1%
1998	232,920	87,802	320,722	72.6%	27.4%	232,000	80,000	290,000	100.4%	109.8%	110.6%
1999	224,565	87,322	311,887	72.0%	28.0%	232,000	90,300	290,000	96.8%	96.7%	107.5%
2000	206,770	22,007	228,777	90.4%	9.6%	232,000	90,300	290,000	89.1%	24.4%	78.9%
2001	173,940	53,585	227,525	76.4%	23.6%	190,400	81,600	238,000	91.4%	65.7%	95.6%
2002	130,453	50,244	180,697	72.2%	27.8%	129,600	–	162,000	100.7%	–	111.5%
2003	141,945	63,217	205,162	69.2%	30.8%	148,200	–	228,000	95.8%	–	90.0%
2004	217,240	125,067	342,307	63.5%	36.5%	250,000	–	514,441	86.9%	–	66.5%
2005	260,120	103,014	363,135	71.6%	28.4%	269,069	95,128	364,197	96.7%	108.3%	99.7%
2006	266,955	94,744	361,699	73.8%	26.2%	269,545	95,297	364,842	99.0%	99.4%	99.1%
2007	217,682	73,550	291,231	74.7%	25.3%	242,591	85,767	328,358	89.7%	85.8%	88.7%
2008	248,395	74,602	322,997	76.9%	23.1%	269,545	95,297	364,842	92.2%	78.3%	88.5%
2009	121,573	57,353	178,926	67.9%	32.1%	135,939	48,061	184,000	89.4%	119.3%	97.2%
2010	170,967	57,016	227,982	75.0%	25.0%	193,935	68,565	262,500	88.2%	83.2%	86.9%
2011	231,261	56,073	287,334	80.5%	19.5%	290,903	102,848	393,751	79.5%	54.5%	73.0%
2012	160,145	47,059	207,204	77.3%	22.7%	186,036	65,773	251,809	86.1%	71.5%	82.3%
2013	233,578	52,249	285,828	81.7%	18.3%	269,745	95,367	365,112	86.6%	54.8%	78.3%
2014	264,141	35,118	299,259	88.3%	11.7%	316,206	111,794	428,000	83.5%	31.4%	69.9%
2015	154,160	39,684	193,844	79.5%	20.5%	325,072	114,928	440,000	47.4%	34.5%	44.1%
2016	262,328	69,743	332,071	79.0%	21.0%	367,553	129,947	497,500	71.4%	53.7%	66.7%
2017	353,842	86,721	440,563	80.3%	19.7%	441,433	156,067	597,500	80.2%	55.6%	73.7%
2018	317,827	95,413	413,240	76.9%	23.1%	441,433	156,067	597,500	72.0%	61.1%	69.2%
2019	316,622	95,013	411,635	76.9%	23.1%	441,433	156,067	597,500	71.7%	60.9%	68.9%
2020	286,788	92,489	379,277	75.6%	24.4%	424,810	104,480	529,290	67.5%	88.5%	71.7%
2021	269,359	57,076	326,435	82.5%	17.5%	369,400	104,480	473,880	72.9%	54.6%	68.9%
2022	291,697	31,671	323,368	90.2%	9.8%	402,646	142,354	545,000	72.4%	22.2%	59.3%
2023	240,588	23,952	264,540	90.9%	9.1%	461,750	163,250	625,000	52.1%	14.7%	42.3%
2024	165,950	4,048	169,999	97.6%	2.4%	410,034	144,966	555,000	40.5%	2.8%	30.6%
2025	239,385	3,913	243,298	98.4%	1.6%	295,520	104,480	400,000	81.0%	3.7%	60.8%

Table A.4. Annual summary of U.S. and Canadian fishery sampling included in this stock assessment by fleet, 1975–2025, though data for 2025 are not yet included in the assessment model. The majority of values are reported as number of hauls but U.S. shoreside and Canadian Shoreside fleets are reported as the number of trips. A dash (–) indicates there was no sampled catch. The number of fish with otoliths sampled per haul has varied over time but is typically small.

Year	U.S. Foreign (hauls)	U.S. Joint- venture (hauls)	U.S. Mother- ship (hauls)	U.S. Combined Mother- ship Catcher- processor (hauls)	U.S. Catcher- processor (hauls)	U.S. Shore- side (trips)	Canada Foreign (hauls)	Canada Joint- venture (hauls)	Canada Shore- side (trips)	Canada Freezer trawlers (hauls)
1975	13	–	–	–	–	–	–	–	–	–
1976	142	–	–	–	–	–	–	–	–	–
1977	320	–	–	–	–	–	–	–	–	–
1978	336	5	–	–	–	–	–	–	–	–
1979	99	17	–	–	–	–	–	–	–	–
1980	191	30	–	–	–	–	–	–	–	–
1981	113	41	–	–	–	–	–	–	–	–
1982	52	118	–	–	–	–	–	–	–	–
1983	–	117	–	–	–	–	–	–	–	–
1984	49	74	–	–	–	–	–	–	–	–
1985	37	19	–	–	–	–	–	–	–	–
1986	88	32	–	–	–	–	–	–	–	–
1987	22	34	–	–	–	–	–	–	–	–
1988	39	42	–	–	–	–	–	3	–	–
1989	–	77	–	–	–	–	–	3	–	–
1990	–	143	–	–	–	15	–	5	–	–
1991	–	–	–	116	–	26	–	18	–	–
1992	–	–	–	164	–	46	–	33	–	–
1993	–	–	–	108	–	36	–	25	3	–
1994	–	–	–	143	–	50	–	41	1	–
1995	–	–	–	61	–	51	–	35	3	–
1996	–	–	–	123	–	35	–	28	1	–
1997	–	–	–	127	–	65	–	27	1	–
1998	–	–	–	149	–	64	–	21	9	–
1999	–	–	–	389	–	80	–	14	26	–
2000	–	–	–	413	–	91	–	25	1	–
2001	–	–	–	429	–	82	–	28	1	–

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Year	U.S. Foreign (hauls)	U.S. Joint- venture (hauls)	U.S. Mother- ship (hauls)	U.S. Combined Mother- ship Catcher- processor (hauls)	U.S. Catcher- processor (hauls)	U.S. Shore- side (trips)	Canada Foreign (hauls)	Canada Joint- venture (hauls)	Canada Shore- side (trips)	Canada Freezer trawlers (hauls)
2002	—	—	—	342	—	71	—	—	36	—
2003	—	—	—	358	—	78	—	—	21	—
2004	—	—	—	381	—	72	—	20	28	—
2005	—	—	—	499	—	58	—	11	31	14
2006	—	—	—	549	—	83	—	21	21	46
2007	—	—	—	524	—	68	—	1	7	29
2008	—	—	324	—	356	63	—	—	20	31
2009	—	—	316	—	278	65	—	—	7	19
2010	—	—	443	—	331	75	—	—	2	9
2011	—	—	481	—	506	81	—	—	1	4
2012	—	—	299	—	332	76	—	—	38	51
2013	—	—	409	—	474	96	—	—	4	61
2014	—	—	423	—	557	68	—	—	28	28
2015	—	—	203	—	431	84	—	—	6	21
2016	—	—	502	—	671	76	—	—	19	40
2017	—	—	353	—	684	112	—	—	68	76
2018	—	—	414	—	569	92	—	—	43	91
2019	—	—	307	—	566	129	—	—	33	103
2020	—	—	192	—	433	99	—	—	32	—
2021	—	—	186	—	409	124	—	—	—	2
2022	—	—	299	—	472	109	—	—	22	16
2023	—	—	127	—	391	76	—	—	12	14
2024	—	—	124	—	319	104	—	—	21	8
2025	—	—	242	—	230	76	—	—	8	2

Table A.5. Recent age-proportion data used in the assessment for the U.S. Catcher-Processor fleet, though data for 2025 are not yet included in the assessment model. Proportions are calculated from numbers of individuals in each age group. Age 15+ is an accumulator group.

Year	Number of fish	Number of hauls	Age (% of total for each year)														
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+
2016	1,995	671	0.40	52.87	2.37	5.57	2.23	31.31	1.56	2.06	0.73	0.20	0.44	0.20	0.00	0.04	0.00
2017	2,026	684	1.75	0.87	50.75	2.36	4.99	3.08	28.79	3.01	2.11	1.17	0.25	0.58	0.17	0.00	0.12
2018	1,670	569	4.58	35.63	1.05	27.44	1.90	2.57	2.83	19.47	2.22	1.05	0.30	0.54	0.15	0.19	0.09
2019	1,685	566	0.00	6.45	26.06	1.43	38.29	1.60	4.00	1.54	17.34	1.20	1.10	0.28	0.14	0.25	0.32
2020	1,281	433	0.00	0.14	9.33	41.91	1.55	29.82	1.72	1.63	1.59	10.41	0.65	1.01	0.07	0.05	0.11
2021	1,206	409	3.88	0.62	2.82	13.37	36.29	1.66	22.87	1.90	1.99	1.64	10.94	1.37	0.43	0.16	0.07
2022	1,269	472	0.89	47.51	1.65	1.90	8.54	19.54	0.74	12.20	1.58	0.74	0.44	2.86	1.08	0.28	0.04
2023	1,277	391	0.69	51.27	24.03	0.78	0.93	3.56	8.87	1.21	4.97	0.59	0.35	0.50	1.91	0.27	0.06
2024	754	319	11.42	9.60	45.72	15.58	0.72	1.54	1.68	6.28	0.77	3.95	0.80	0.62	0.33	0.57	0.44
2025	581	230	12.54	24.35	2.93	25.20	19.02	0.94	0.66	2.57	4.52	1.04	2.87	0.48	0.62	0.29	1.98

Table A.6. Recent age-proportion data used in the assessment for the U.S. Mothership fleet, though data for 2025 are not yet included in the assessment model. Proportions are calculated from numbers of individuals in each age group. Age 15+ is an accumulator group.

Year	Number of fish	Number of hauls	Age (% of total for each year)														
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+
2016	1,495	502	0.53	59.25	1.45	5.10	2.44	26.82	1.54	1.92	0.38	0.32	0.09	0.15	0.00	0.00	0.00
2017	1,054	353	7.78	0.77	51.20	2.21	3.41	1.28	27.73	1.88	1.96	0.49	0.08	0.81	0.19	0.16	0.06
2018	1,230	414	16.95	25.30	1.18	28.83	1.14	2.28	1.70	16.82	2.47	1.24	0.74	0.32	0.48	0.49	0.05
2019	903	307	0.00	14.98	20.59	0.97	36.30	1.33	4.12	1.53	16.62	1.47	1.04	0.42	0.48	0.14	0.01
2020	568	192	0.00	0.00	8.62	40.11	2.40	28.62	1.49	2.06	2.51	11.89	1.12	0.80	0.39	0.00	0.00
2021	545	186	0.00	0.43	1.78	11.57	37.92	2.18	22.34	1.27	1.98	2.77	13.83	2.40	0.67	0.21	0.67
2022	840	299	1.45	42.64	1.97	2.83	6.91	19.41	1.07	14.51	1.82	0.47	0.52	3.82	1.53	0.43	0.62
2023	448	127	2.28	39.59	33.18	1.15	1.21	5.46	7.49	0.67	5.19	0.70	0.25	0.49	2.06	0.29	0.00
2024	267	124	4.83	14.20	58.93	11.41	0.47	0.00	1.76	2.97	0.00	2.79	0.52	0.20	0.00	0.23	1.71
2025	663	242	3.42	58.90	7.46	17.84	7.06	0.38	0.28	0.37	2.32	0.12	1.00	0.43	0.20	0.07	0.15

Table A.7. Recent age-proportion data used in the assessment for the U.S. shoreside fleet, though data for 2025 are not yet included in the assessment model. Proportions are calculated from numbers of individuals in each age group. Age 15+ is an accumulator group.

Year	Number of fish	Number of trips	Age (% of total for each year)														
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+
2016	1,518	76	0.11	65.44	1.41	3.25	1.55	22.03	1.60	2.70	0.72	0.29	0.31	0.26	0.14	0.10	0.08
2017	2,235	112	3.68	0.71	35.37	2.63	3.66	2.50	43.03	2.89	2.12	1.66	0.64	0.53	0.27	0.11	0.20
2018	1,834	92	7.72	27.85	1.75	31.45	1.24	2.40	2.61	19.08	2.65	1.32	0.86	0.49	0.40	0.15	0.05
2019	2,566	129	0.00	15.79	22.48	0.93	32.20	1.86	3.30	1.74	16.71	1.28	1.61	0.90	0.54	0.31	0.37
2020	1,974	99	0.00	0.02	8.34	34.50	1.35	32.07	1.24	2.29	1.57	15.88	1.06	0.88	0.43	0.06	0.32
2021	2,480	124	0.17	0.26	1.97	12.69	34.48	2.73	25.93	1.92	2.80	2.08	11.12	2.27	0.85	0.22	0.50
2022	2,180	109	0.50	15.14	1.46	2.86	11.07	31.21	2.36	18.84	2.40	1.59	1.67	7.63	2.07	0.67	0.51
2023	1,596	76	0.11	17.99	26.23	1.82	3.34	8.43	19.32	2.03	11.38	1.87	0.73	1.22	3.99	0.72	0.82
2024	2,178	104	7.78	2.79	30.52	17.98	1.64	2.18	6.30	12.77	1.33	8.85	1.25	1.06	0.83	3.21	1.51
2025	1,629	76	1.78	39.84	5.68	22.53	12.28	1.18	1.00	1.88	6.53	0.83	3.08	0.88	0.47	0.26	1.78

Table A.8. Recent age-proportion data used in the assessment for the Canadian Shoreside fleet, though data for 2025 are not yet included in the assessment model. Proportions are calculated from numbers of individuals in each age group. Age 15+ is an accumulator group.

Year	Number of fish	Number of trips	Age (% of total for each year)														
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+
2016	188	19	0.00	4.66	0.81	7.51	3.91	62.24	5.83	7.35	1.54	2.10	0.00	1.22	0.91	0.27	1.65
2017	680	68	6.95	0.34	7.85	1.72	3.01	7.28	48.08	13.21	6.94	1.33	1.25	1.20	0.14	0.15	0.55
2018	466	43	0.50	5.15	1.91	22.50	1.23	4.48	5.93	35.33	12.44	4.43	2.61	1.05	0.96	1.23	0.24
2019	296	33	0.00	13.24	11.41	2.87	30.27	1.90	4.36	2.70	26.37	2.28	3.26	0.83	0.51	0.00	0.00
2020	1,438	32	0.00	0.04	9.59	19.80	1.37	30.16	2.71	3.49	2.56	24.07	2.86	2.11	0.22	0.48	0.54
2022	596	22	0.00	0.00	0.13	1.42	13.76	22.91	6.59	17.47	4.75	4.29	4.52	13.98	5.88	2.41	1.88
2023	413	12	0.00	0.31	5.16	1.37	2.64	17.33	31.55	6.31	13.18	1.35	2.87	3.12	9.43	3.12	2.26
2024	819	21	0.00	0.00	0.04	2.78	1.68	3.46	18.00	30.80	3.11	15.91	2.75	3.38	2.24	11.71	4.14
2025	198	8	0.00	0.38	0.24	0.97	12.20	5.12	6.36	24.12	21.46	3.36	11.62	2.16	1.35	2.45	8.21

Table A.9. Recent age-proportion data used in the assessment for the Canadian Freezer trawler fleet, though data for 2025 are not yet included in the assessment model. Proportions are calculated from numbers of individuals in each age group. Age 15+ is an accumulator group.

Year	Number of fish	Number of hauls	Age (% of total for each year)														
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+
2016	352	40	0.51	4.54	0.77	2.22	9.06	64.28	6.93	6.87	2.00	0.75	0.16	0.70	0.42	0.00	0.79
2017	760	76	0.00	0.58	7.34	2.42	5.49	5.07	50.05	12.21	9.71	2.37	2.49	1.38	0.21	0.19	0.50
2018	1,225	91	0.10	4.71	0.72	17.63	2.47	3.96	5.14	45.59	9.45	5.25	2.37	1.15	0.65	0.56	0.26
2019	901	103	0.04	18.00	15.03	3.66	19.28	2.76	3.95	4.60	23.06	5.35	2.39	1.14	0.38	0.37	0.00
2021	100	2	0.00	0.00	0.00	17.01	24.00	6.99	27.50	2.84	2.84	9.34	5.50	1.17	0.00	2.84	0.00
2022	421	16	0.00	0.00	0.00	0.24	6.20	22.15	8.02	17.76	7.55	4.38	6.08	16.52	7.45	2.87	0.76
2023	369	14	0.00	0.00	0.27	0.40	3.26	15.99	21.59	5.06	19.62	6.11	3.20	7.55	13.89	1.95	1.10
2024	390	8	0.00	0.00	0.17	1.15	3.16	3.00	14.92	29.92	12.95	18.87	2.99	1.61	1.32	6.28	3.67
2025	50	2	0.00	0.00	0.00	1.24	16.00	1.24	2.48	11.82	38.41	2.48	15.54	0.00	0.00	2.70	8.11

B DATA FIGURES

This appendix contains empirical data figures only. There are no model output figures included. The 2026 assessment is a simple projection model which includes only the total 2025 catch as new data. The figures presented here contain historical data as well as the data for 2025, even though only catch was updated in the modelling for this year. All the 2025 data presented here will be used in future assessment years' model runs.

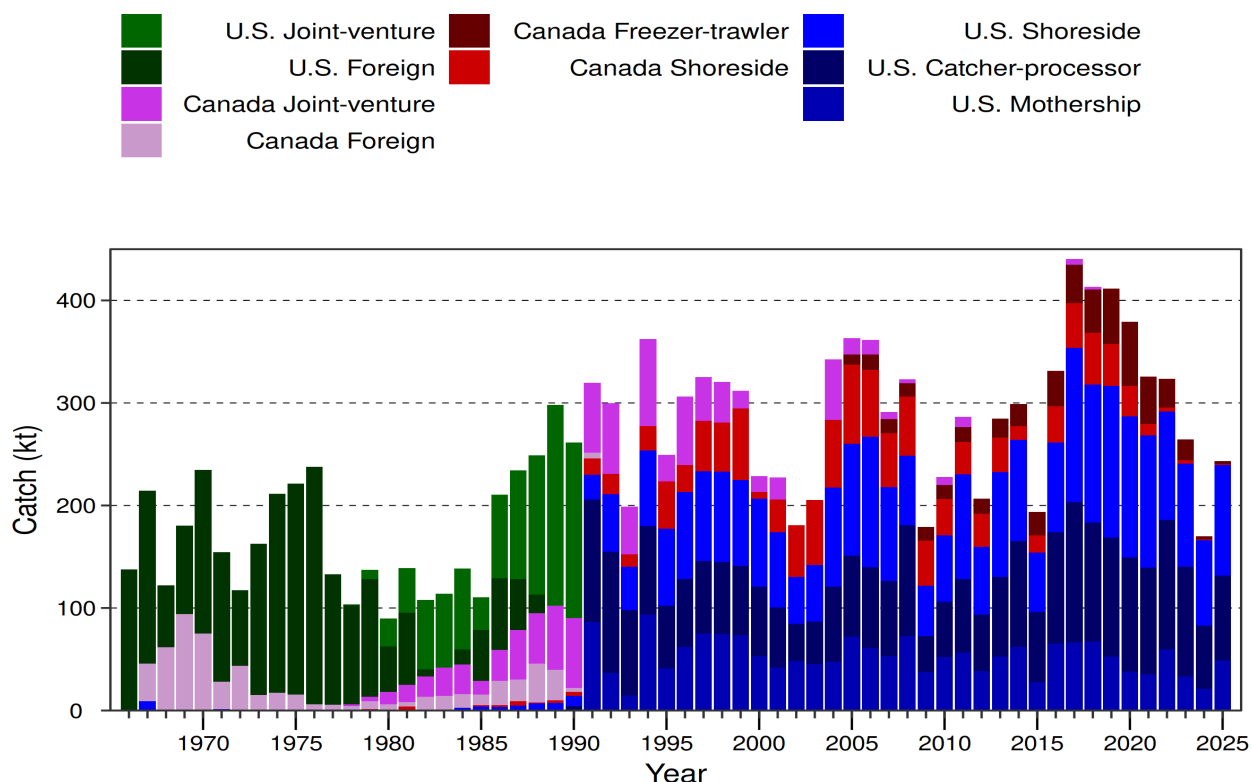


Figure B.1. Total Pacific Hake catch used in the assessment by sector, 1966–2025. U.S. tribal catches are included in the sectors where they are represented.

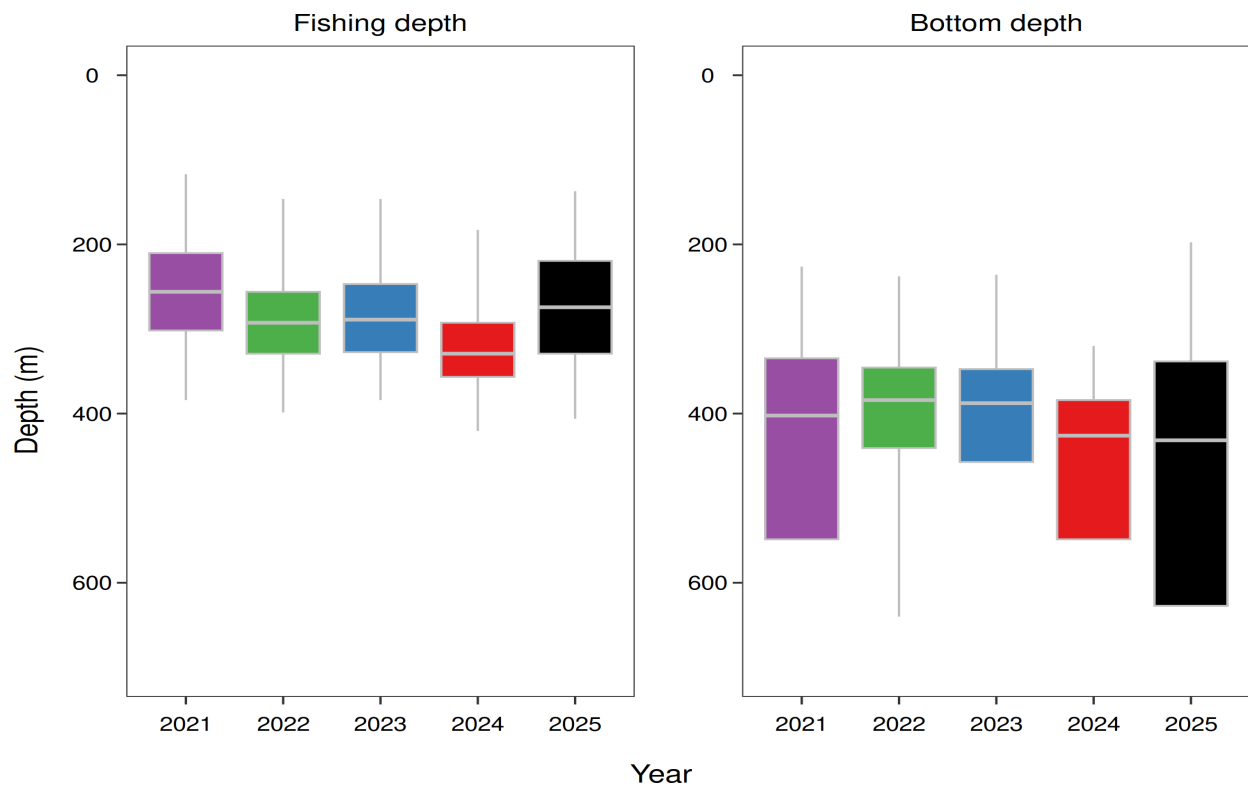


Figure B.2. Distribution of fishing depths (left) and bottom depths (right), in meters, of hauls targeting Pacific Hake in the U.S. Catcher-Processor and Mothership sectors from 2021–2025. Horizontal lines in each box represent the median depth and boxes encompass the middle 50% of the data. Whiskers encompass the 95% quantiles.

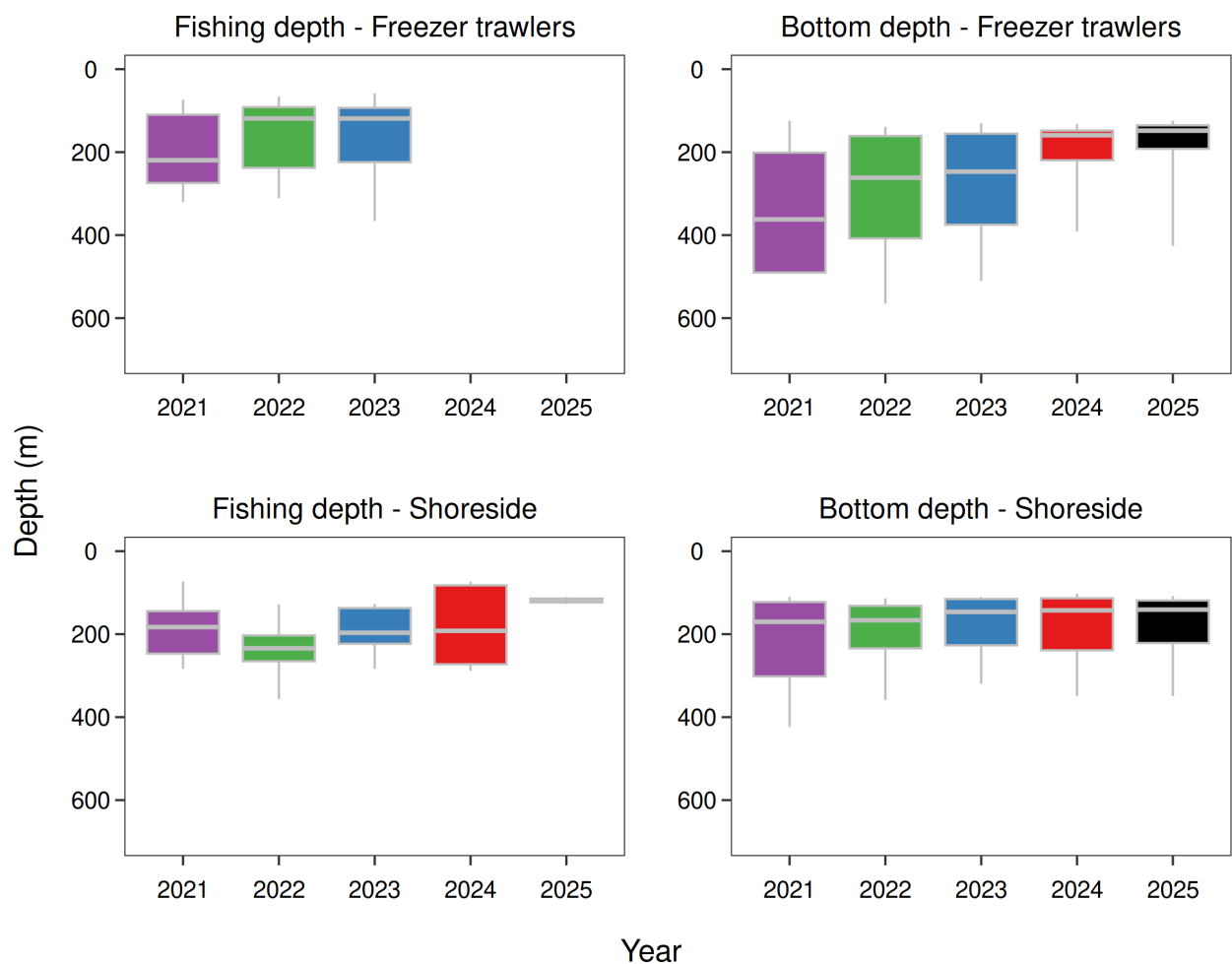


Figure B.3. Distribution of fishing depths (left) and bottom depths (right), in meters, of hauls targeting Pacific Hake in the Canadian fleets from 2021–2025. Horizontal lines in each box represent the median depth and boxes encompass the middle 50% of the data. Whiskers encompass the 95% quantiles.

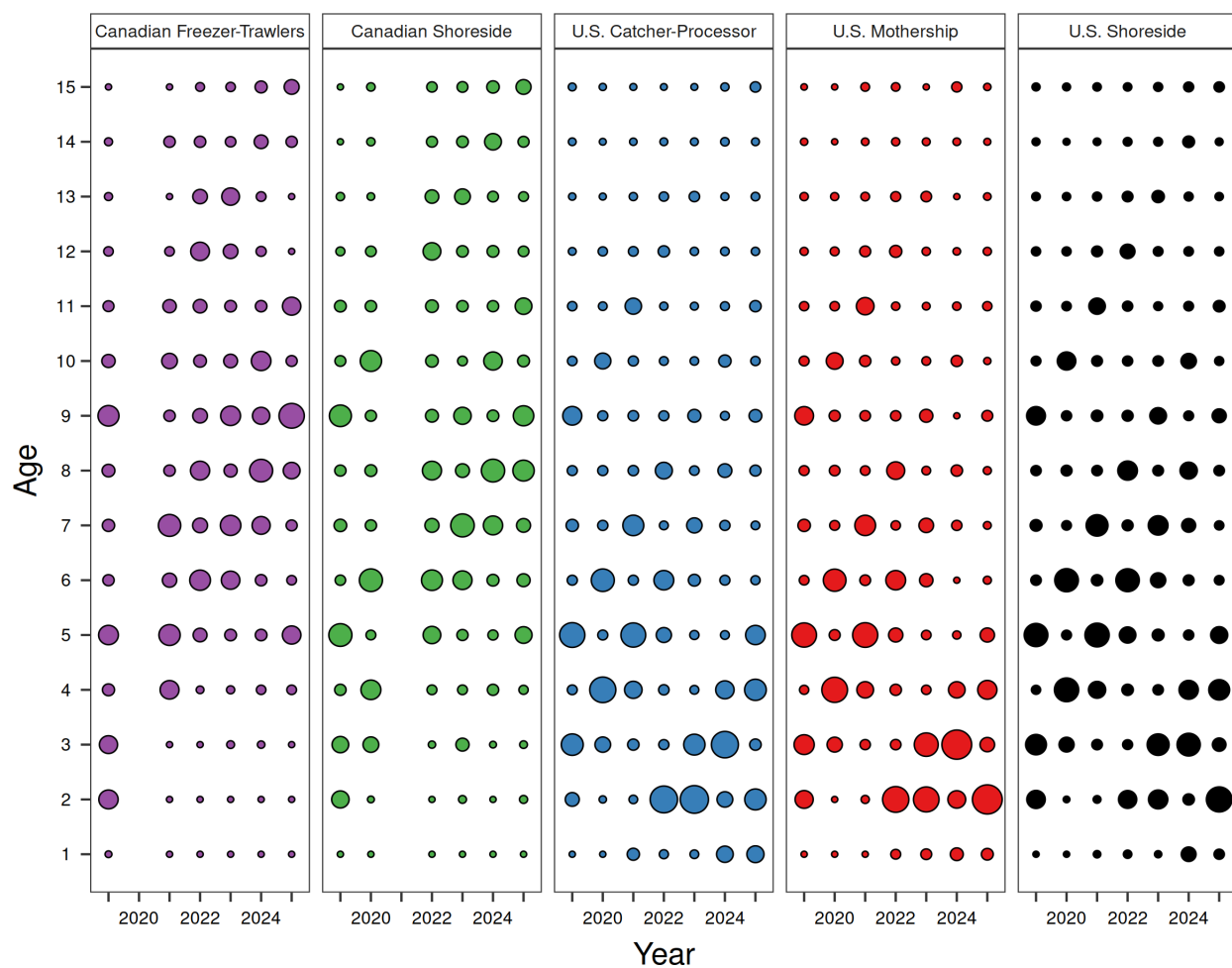


Figure B.4. Recent age-proportion data for U.S. and Canadian fleets. Proportions are calculated from numbers of individuals in each age group. Age 15+ is an accumulator group.

C GLOSSARY OF TERMS USED IN THIS DOCUMENT

40:10 adjustment: A reduction in the overall total allowable catch that is triggered when the female spawning biomass falls below 40% of its unfished equilibrium level. This adjustment reduces the total allowable catch on a straight-line basis from the 40% level such that the total allowable catch would equal zero when the biomass is at 10% of its unfished equilibrium level. This is one component of the default harvest policy.

Acceptable biological catch (ABC): The acceptable biological catch is a scientific calculation of the sustainable harvest level of a fishery used historically to set the upper limit for fishery removals by the Pacific Fishery Management Council. It is calculated by applying the estimated (or proxy) harvest rate that produces the maximum sustainable yield from the estimated vulnerable biomass. For Pacific Hake, the calculation of the acceptable biological catch and application of the 40:10 adjustment is now replaced with the default harvest rate and the total allowable catch.

Adjusted: A term used to describe the total allowable catch or allocations that account for carryovers of uncaught catch from previous years.

Advisory Panel (AP): The Advisory Panel on Pacific Hake established by the Agreement.

Agreement ('Treaty'): The Agreement between the government of the United States and the government of Canada on Pacific Hake, signed in Seattle, Washington, on November 21, 2003 and entered into force June 25, 2008.

Alaska Fisheries Science Center (AFSC): One of six regional NOAA Fisheries Science Centers, primarily in Seattle, Washington but also present throughout Alaska.

B_0 : Unfished equilibrium female spawning biomass.

$B_{10\%}$: The level of female spawning biomass corresponding to 10% of unfished equilibrium female spawning biomass, i.e., $B_{10\%} = 0.1B_0$. This is the level below which the calculated total allowable catch is set to 0, based on the 40:10 adjustment.

$B_{40\%}$: The level of female spawning biomass corresponding to 40% of unfished equilibrium female spawning biomass, i.e., $B_{40\%} = 0.4B_0$. This is the level below which the total allowable catch is decreased from the value associated with $F_{SPR=40\%}$, based on the 40:10 adjustment.

B_{MSY} : The estimated female spawning biomass which theoretically would produce the maximum sustainable yield under equilibrium fishing conditions (constant fishing and average recruitment in every year).

Backscatter: The scattering by a target back in the direction of an acoustic source. Specifically, the Nautical Area Scattering Coefficient (a measure of scattering per area) is frequently referred to as backscatter.

Benchmark spawning potential ratio ($B_{SPR=40\%}$): The spawning potential ratio of 40%, where 40% relates to the default harvest rate of $F_{SPR=40\%}$ specified in the Agree-

ment. Even under equilibrium conditions, $F_{\text{SPR}=40\%}$ would not necessarily result in a female spawning biomass of $B_{40\%}$ because $F_{\text{SPR}=40\%}$ is defined in terms of the spawning potential ratio that depends on the female spawning biomass per recruit.

California Current Ecosystem: The waters of the continental shelf and slope off the west coast of North America, commonly referring to the area from Central California to Southern British Columbia.

Carryover: If at the end of the year, there are unharvested allocations, then there are provisions for some of these unharvested allocations to be carried over into the next year's allocation process. The Agreement states that "If, in any year, a Party's catch is less than its individual total allowable catch, an amount equal to the shortfall shall be added to its individual total allowable catch in the following year, unless otherwise recommended by the Joint Management Committee. Adjustments under this sub-paragraph shall in no case exceed 15 percent of a Party's unadjusted individual total allowable catch for the year in which the shortfall occurred."

Catchability (q): The parameter defining the proportionality between a relative index of abundance (often a fishery-independent survey) and the estimated abundance available to that survey (as modified by selectivity) in the assessment model.

Catch-per-unit-effort (CPUE): A raw or, frequently, standardized and model-based metric of fishing success based on the catch and relative effort expended to generate that catch. Catch-per-unit-effort is often used as an index of abundance in the absence of fishery-independent indices and/or where the two types of indices are believed to be proportional.

Catch target: A general term used to describe the catch value used for management. Depending on the context, this may be a limit rather than a target and may be equal to a total allowable catch, an acceptable biological catch, the median result of applying the default harvest policy, or some other number. The Joint Technical Committee welcomes input from the Joint Management Committee on the best terminology to use for these quantities.

Closed-loop simulation: A subset of a management strategy evaluation that iteratively simulates a population using an operating model, generates data from that population and passes it to an estimation method, uses the estimation method and a management strategy to provide management advice, which then feeds back into the operating model to simulate an additional fixed set of time before repeating this process.

Coefficient of variation (CV): A measure of uncertainty defined as the standard deviation divided by the mean.

Cohort: A group of fish born in the same year. Also see recruitment and year-class.

Constant catch: A catch scenario used for forecasting in which the same catch is used in successive years.

Default harvest policy (rate): The application of $F_{\text{SPR}=40\%}$ with the 40:10 adjustment. Having considered any advice provided by the Joint Technical Committee, Scientific Review Group, or Advisory Panel, the Joint Management Committee may recommend a different harvest rate if the scientific evidence demonstrates that a different rate is necessary to sustain the resource.

Department of Fisheries and Oceans (DFO) Canada: See Fisheries and Oceans Canada.

Depletion: Prior to the 2015 assessment, depletion was used instead of relative spawning biomass. ‘Relative depletion’ was also used.

El Niño: Abnormally warm ocean climate conditions in the California Current Ecosystem as a result of broad changes in the Eastern Pacific Ocean across the eastern coast of Latin America (centered on Peru) often around the end of the calendar year.

Exploitation fraction: A metric of fishing intensity that represents the total annual catch divided by the estimated population biomass over a range of ages assumed to be vulnerable to the fishery (set to ages 2+ in this assessment; note that in some previous assessments it was 3+). This value is not equivalent to the instantaneous rate of fishing mortality or the spawning potential ratio.

$F_{\text{SPR}=40\%}$: The rate of fishing mortality estimated to give a spawning potential ratio of 40%. Therefore, by definition this satisfies

$$0.4 = \frac{\text{spawning biomass per recruit with } F_{40\%}}{\text{spawning biomass per recruit with no fishing}}, \quad (\text{C.1})$$

and $\text{SPR}(F_{40\%}) = 40\%$. The 40% value is specified in the Agreement.

$F_{\text{SPR}=40\%}$ –40:10 harvest policy: The default harvest policy.

Female spawning biomass: The biomass of mature female fish at the beginning of the year. Sometimes abbreviated to spawning biomass.

Fisheries and Oceans Canada (DFO): Federal organization that delivers programs and services to support sustainable use and development of Canada’s waterways and aquatic resources. Was previously called Department of Fisheries and Oceans.

Fishing intensity: A measure of the magnitude of fishing, defined for a fishing rate (F) as:

$$\text{fishing intensity for } F = 1 - \text{SPR}(F), \quad (\text{C.2})$$

where $\text{SPR}(F)$ is the spawning potential ratio for the value of F accumulated over the entire year. It is often given as a percentage. Relative fishing intensity (Figures C.1 and C.2) is the fishing intensity relative to that at the SPR fishing rate $F_{\text{SPR}=40\%}$, where $F_{\text{SPR}=40\%}$ is the F that gives an SPR of 40% such that, by

definition, $\text{SPR}(F_{40\%}) = 40\%$ (the benchmark spawning ratio). Therefore

$$\text{relative fishing intensity for } F = \frac{1 - \text{SPR}(F)}{1 - \text{SPR}(F_{40\%})} \quad (\text{C.3})$$

$$= \frac{1 - \text{SPR}(F)}{1 - 0.4} \quad (\text{C.4})$$

$$= \frac{1 - \text{SPR}(F)}{0.6}. \quad (\text{C.5})$$

For brevity we use $F_{\text{SPR}=40\%} = \text{SPR}(F_{40\%})$ in the text. Although this simply equals 40%, it can be helpful to explicitly write:

$$\text{relative fishing intensity for } F = \frac{1 - \text{SPR}(F)}{1 - \text{SPR}_{40\%}}. \quad (\text{C.6})$$

Fishing mortality rate or instantaneous rate of fishing mortality (F): A metric of fishing intensity that is usually reported in relation to the most highly selected ages(s) or length(s), or occasionally as an average over an age range that is vulnerable to the fishery. Because it is an instantaneous rate operating simultaneously with natural mortality, it is not equivalent to exploitation fraction, percent annual removal, or the spawning potential ratio.

F_{MSY} : The rate of fishing mortality estimated to produce the maximum sustainable yield from the population.

Harvest strategy: A formal system for managing a fishery that includes the elements shown in Figure A.1 of Taylor et al. (2015).

Harvest control rule: A process for determining an acceptable biological catch from a stock assessment. Also see default harvest policy.

Integrated West Coast Pelagic Survey (IWCPs): A fisheries-independent survey that began in 2025 to inform both Pacific hake and several coastal pelagic species in U.S. waters. The IWCPs continues to collect acoustic, biological, and ecosystem data, as did previous species-independent acoustic-trawl surveys. The IWCPs utilizes a contemporary multifunction net designed cooperatively with industry that can complete mid-water and surface trawl tows. The IWCPs and the Department of Fisheries and Oceans, Canada, acoustic-trawl survey are carefully coordinated and joined to develop the Joint U.S. and Canadian Integrated Acoustic and Trawl Survey for Pacific hake.

Joint Management Committee (JMC): The Joint Management Committee is established by the Agreement.

Joint Technical Committee (JTC): The Joint Technical Committee is established by the Agreement. The formal name is 'Joint Technical Committee of the Pacific Hake/Whiting Agreement Between the Governments of the United States and Canada'.

Kilotonne (kt). Metric abbreviation for 1,000 metric tonnes.

Logistic transformation: A mathematical transformation used to translate between numbers bounded within some range to numbers on the real line ($-\infty$ to $+\infty$).

Magnuson–Stevens Fishery Conservation and Management Act (MSFCMA): The MSFCMA, sometimes known as the ‘Magnuson–Stevens Act’, established the 200-mile fishery conservation zone, the regional fishery management council system, and other provisions of U.S. marine fishery law.

Management Strategy Evaluation (MSE): A formal process for evaluating harvest strategies.

Markov chain Monte Carlo (MCMC): A numerical method used to sample from the posterior distribution of parameters and derived quantities in a Bayesian analysis. It is more computationally intensive than computing the maximum likelihood estimate but provides a more accurate depiction of parameter uncertainty. See Stewart et al. (2013) for a discussion of issues related to differences between Markov chain Monte Carlo and maximum likelihood estimation.

Maximum sustainable yield (MSY): An estimate of the largest sustainable annual catch that can be continuously taken over a long period of time from a population under equilibrium ecological and environmental conditions.

Megatonne (Mt): Metric abbreviation for 1,000,000 metric tonnes.

Metric tonne (t): A unit of mass equal to 1,000 kilograms or 2,204.62 pounds. Some previous stock assessments used the abbreviation ‘mt’.

National Marine Fisheries Service (NMFS): See NOAA Fisheries.

No-U-Turn Sampler (NUTS): An advanced Hamiltonian Bayesian Markov chain Monte Carlo sampling algorithm used to efficiently create posterior distributions and used in Pacific Hake Bayesian assessments beginning in 2021.

NOAA Fisheries: The division of the United States National Oceanic and Atmospheric Administration (NOAA) responsible for conservation and management of off-shore fisheries (and inland salmon). This is also known as the National Marine Fisheries Service (NMFS), and both names are commonly used at this time.

North Pacific Groundfish and Halibut Observer Program Database (NORPAC): A database that stores data collected at sea by U.S. fishery observers.

Northwest Fisheries Science Center (NWFSC): One of six regional NOAA Fisheries Science Centers, primarily in Seattle, Washington but also in various locations in Oregon and Washington.

Not available (NA): Something that is not available, e.g., an entry in a table.

Operating model (OM): A model used to simulate data for use in the management strategy evaluation. The operating model includes components for the population and

fishery dynamics, as well as the simulation of the data sampling process, potentially including observation error. Cases in the management strategy evaluation represent alternative configurations of the operating model.

Pacific Biological Station (PBS): The Pacific Biological Station of Fisheries and Oceans Canada located in Nanaimo, British Columbia.

Pacific Fisheries Information Network (PacFIN): A database that provides a central repository for commercial fishery information from Washington, Oregon, and California.

Pacific Fishery Management Council (PFMC): The U.S. organization under which historical stock assessments for Pacific Hake were conducted.

Pacific Hake: Common name for *Merluccius productus*, the species whose offshore population in the waters of the United States and Canada is subject of this assessment.

Pacific whiting: An alternative name for Pacific Hake commonly used in the United States.

Posterior distribution: The probability distribution for parameters or derived quantities from a Bayesian model representing the result of the prior probability distributions being updated by the observed data via the likelihood equation. For stock assessments, posterior distributions are approximated via numerical methods; one frequently employed method is Markov chain Monte Carlo sampling.

Prior distribution: A probability distribution for a parameter in a Bayesian analysis that represents the information available before evaluating the observed data via the likelihood equation. For some parameters, uninformative priors can be constructed that allow the data to dominate the posterior distribution. For other parameters, informative priors can be constructed based on auxiliary information and/or expert knowledge or opinions.

R_0 : Estimated annual recruitment at unfished equilibrium.

Random walk Metropolis Hastings (rwMH): Bayesian Markov chain Monte Carlo sampling algorithm used to create posterior distributions used in Pacific Hake Bayesian stock assessment models prior to 2021.

Recruits/recruitment: the estimated number of new members in a fish population born in the same age. In this assessment, recruitment is reported at age 0. See cohort and year-class.

Recruitment deviation: The offset of the recruitment in a given year relative to the stock–recruitment relationship; values occur on a logarithmic scale and are relative to the expected recruitment at a given female spawning biomass.

Relative fishing intensity: See fishing intensity.

Relative spawning biomass: The ratio of the beginning-of-the-year female spawning biomass to the unfished equilibrium female spawning biomass (B_0). Thus, lower values are associated with fewer mature female fish. This term was introduced in the 2015 stock assessment as a replacement for ‘depletion’.

Scientific Review Group (SRG): The Scientific Review Group established by the Agreement.

Scientific and Statistical Committee (SSC): The scientific advisory committee to the Pacific Fishery Management Council. The Magnuson–Stevens Fishery Conservation and Management Act requires that each council maintain a Scientific and Statistical Committee to assist in gathering and analyzing statistical, biological, ecological, economic, social, and other scientific information that is relevant to the management of the Council.

Simulation: A model evaluation under a particular state of nature, including combinations of parameters controlling stock productivity, stock status, and the time series of recruitment deviations. In this assessment, there are 8,000 simulations used to characterize alternative states of nature, each of which are based on a sample from the posterior distribution of the parameters, as calculated using Markov chain Monte Carlo, for a particular model (e.g., the base model).

Spawning biomass: Abbreviated term for female spawning biomass.

Spawning biomass per recruit: The expected lifetime contribution of an age-0 recruit, calculated as the sum across all ages of the product of spawning biomass at each age and the probability of surviving to that age. See Figure C.2 for a graphical demonstration of the calculation of this value, which is found in both numerator and denominator of the spawning potential ratio.

Spawning potential ratio (SPR): The ratio of the spawning biomass per recruit under a given level of fishing to the estimated spawning biomass per recruit in the absence of fishing. Often expressed as a percentage, it achieves a value of 100% in the absence of fishing and declines toward zero as fishing intensity increases (Figure C.2).

Standard deviation (sd): A measure of variability about the mean within a sample.

Steepness (h): A parameter of the stock–recruitment relationship representing the proportion of R_0 expected (on average) when the female spawning biomass is reduced to 20% of B_0 (i.e., when relative spawning biomass is equal to 20%).

Stock Synthesis (SS3): The age-structured stock assessment model applied in this stock assessment.

Target strength (TS): The amount of backscatter from an individual acoustic target.

Total allowable catch (TAC): The maximum fishery removal under the terms of the Agreement.

U.S./Canadian allocation: The division of the total allowable catch as 73.88% for the U.S. share and 26.12% for the Canadian share.

Vulnerable biomass: The demographic portion of the population available for harvest by the fishery.

Year-class: A group of fish born in the same year. See ‘cohort’ and ‘recruitment’.

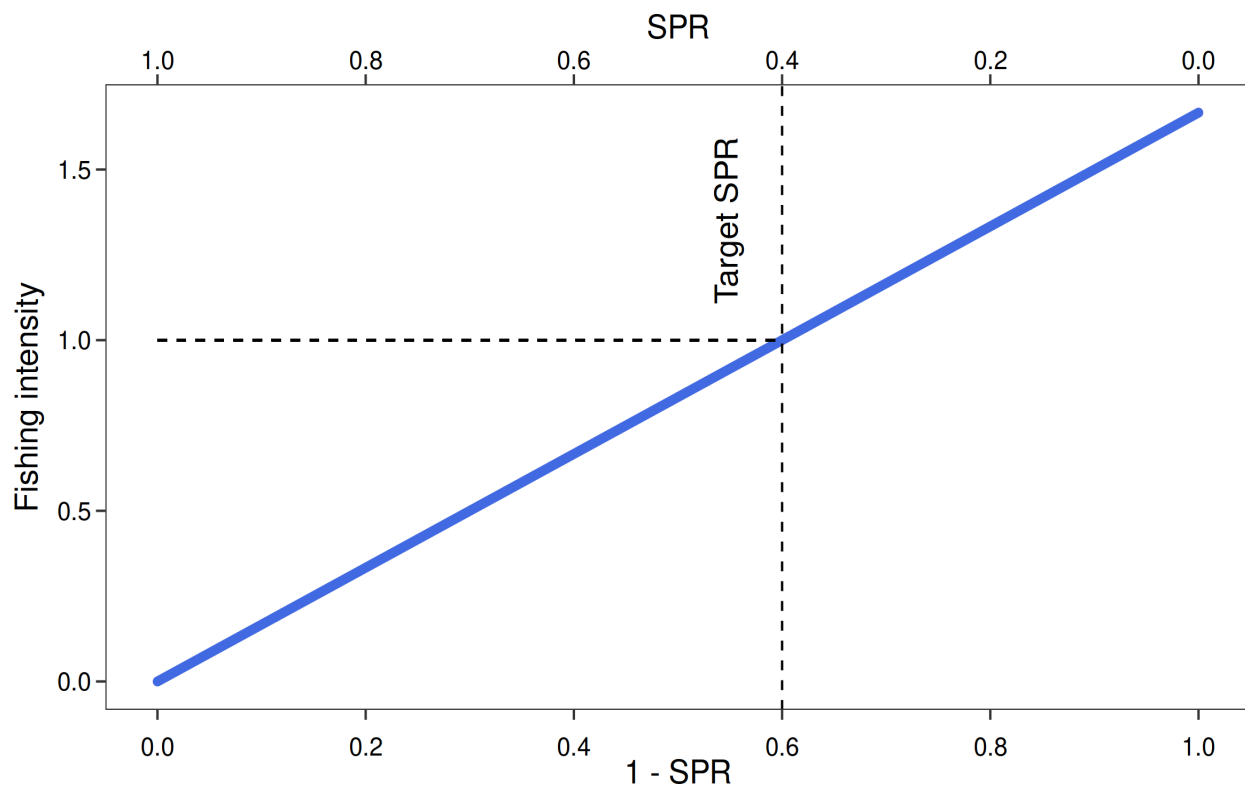


Figure C.1. Fishing intensity as a function of the spawning potential ratio (SPR; top axis) and 1-SPR (bottom axis); given the benchmark SPR of 40%, the solid blue line is simply $1/0.6$, as shown in equation (C.3).

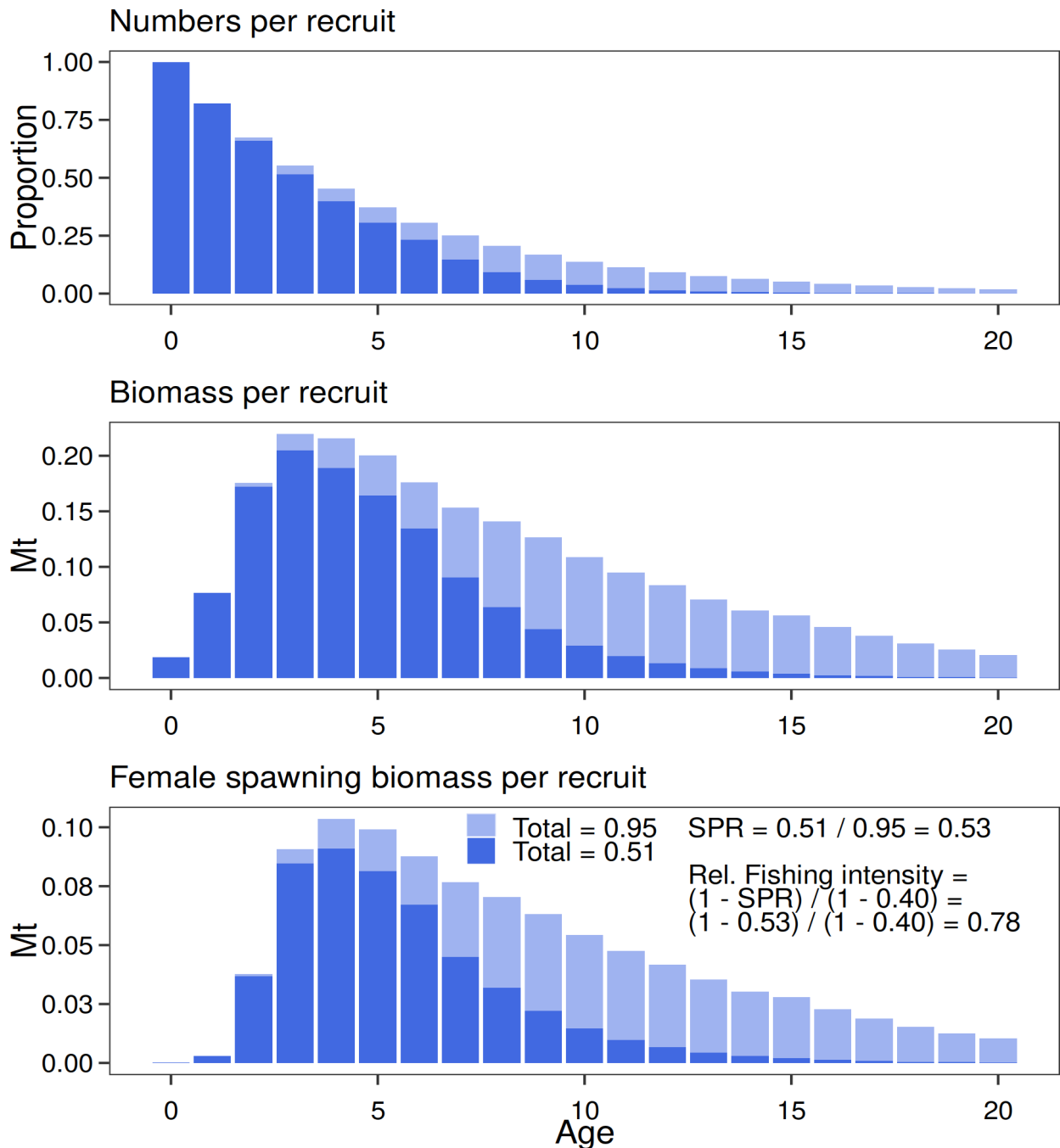


Figure C.2. Illustration of the spawning potential ratio (SPR) calculation based on the combination of maturity and fecundity used in the model, using the Markov chain Monte Carlo (MCMC) estimates of natural mortality, selectivity, and fishing mortality in the final year of the base model used in the 2025 assessment. The light blue bars represent unfished values, the dark blue bars represent fished values. The numbers per recruit are scaled to be 1 at age-0, and biomass values are in millions of metric tonnes.

D REPORT OF THE 2025 PACIFIC HAKE FISHERY IN THE UNITED STATES

Prepared by the United States Advisory Panel on January 23, 2026 for inclusion in this assessment document.

2025 U.S. Sector Allocations and Catch

Based on information provided by U.S. Advisory Panel members and data from the Pacific Fisheries Information Network (PacFIN), total U.S. harvest in the 2025 whiting fishery was **239,206 t** (see Table D.1), which is about 81% of the U.S. allocation of 295,520 t. Compared to the 2024 season (165,582 t), total U.S. catch was up 45% in 2025.

Table D.1. 2025 U.S. whiting allocations and catch (tonnes, t). Tribal catch is estimated (SS sector), and MS catch was provided by the MS cooperative. Dashes represent unknown values. Tribal reapportionment occurred twice, on September 16 and September 29, 2025.

U.S. Sector	Initial Allocation	Allocation After Reapportionment	Catch (t)	Utilization (%)
Tribal	51,716	10,000	526	5
Research	750	750	–	–
Shoreside (SS)	102,083	119,603	107,185	90
Mothership (MS)	58,333	68,345	48,938	72
Catcher-Processor (CP)	82,638	96,822	82,557	85
Total	295,520	295,520	239,206	81

Sources for Table D.1 are from the [Federal register](#) and [PacFIN Whiting Report](#).

Shoreside (SS)

Similar to 2024, the 2025 shoreside fishery started out slow (no fish caught in the first two weeks of May). Fishing steadily picked up as the season progressed and continued all the way until the end of November, although effort began to drop off in late October due to weather and some boats heading to shipyard. Summer fishing for the SS sector was reported to be very good, as boats saw large schools of approximately 500g fish and were able to fill up quickly. In the fall SS fishery, (Sept. through Nov.), fish were caught from Coos Bay all the way up to Juan de Fuca. Fish size increased over the season and was reported by shoreside processors to average 510-550g across the year.

Total catch in the 2025 SS Fishery was 107,185 t, which was about 90% of the total SS allocation. Several SS catcher boats did run up against their 2025 individual allocation and were either forced to quit early in the season or lease additional hake quota to continue fishing.

Mothership (MS)

The 2025 mothership fishery was much improved compared to previous years. The MS fishery began mid-June and ended October 23, 2025. Fish availability was better and generally consistent throughout the year. Over the course of the year, MS fishing activity

was distributed primarily south of Newport, OR. Improved utilization by the MS sector in 2025 is largely due to the consistent availability of MS platforms on the fishing grounds throughout the fishing year, as the two MS platforms that participated in the 2025 fishery remained on the grounds throughout the summer. The MS catcher boats were able to turn several trips with good, clean fishing, good fish size (averaging about 417g), and great catch per unit effort (CPUE).

Total catch in the 2025 MS fishery was 48,938 t, which was about 72% of the MS allocation. Several MS catcher boats did run up against their 2025 individual allocation and were either forced to quit early in the season or lease additional hake quota to continue fishing.

Catcher Processor (CP)

Overall fishing conditions for the CP sector were improved compared to last year, with better fish availability north of the California/Oregon border. Nearly all fishing activity took place off the Oregon Coast, distributed from the Columbia River to the California border. The CP fishery began in mid-May, and all CPs completed fishing operations by early November.

Total catch in the 2025 CP fishery was 82,557 t, which was about 85% of the CP allocation. The primary reason the CP sector did not catch 100% of its allocation is due to a loss of two CP vessels from one company over the past several years. Its important to note two of the three CP companies that participated in the 2025 CP fishery caught nearly all of their allocations.

E REPORT OF THE 2025 PACIFIC HAKE FISHERY IN CANADA

Prepared by the Canadian Advisory Panel on January 23, 2026 for inclusion in this assessment document.

The 2025 Canadian Offshore Pacific Hake fishery catch was the second lowest recorded since the fishery began in the late 1970s, with 3,988 mt caught at today's date. This is a negligible increase from the 3,105 mt caught in 2024 when compared with the overall coastwide total catch.

The reports from the fishermen were consistent; fishing was very poor as evident in the extremely low catch. The fleet reported no body of Hake on the Canadian Zone that allowed for a fishery to take place. What little catch occurred was in the traditional grounds off Ucluelet. Frozen at-sea (FAS) activity was largely one vessel only, and they searched the traditional area off Winter Harbour in the shallows and off the edge, but no fish was found. The fish that were encountered were again large older +1,000gr. The bycatch was primarily Pollock. The market for Hake was strong again in 2025, but the catches were not there to supply the fleet or the plants. The FAS Vessels accounted for an estimated 95% of the catch in 2025.

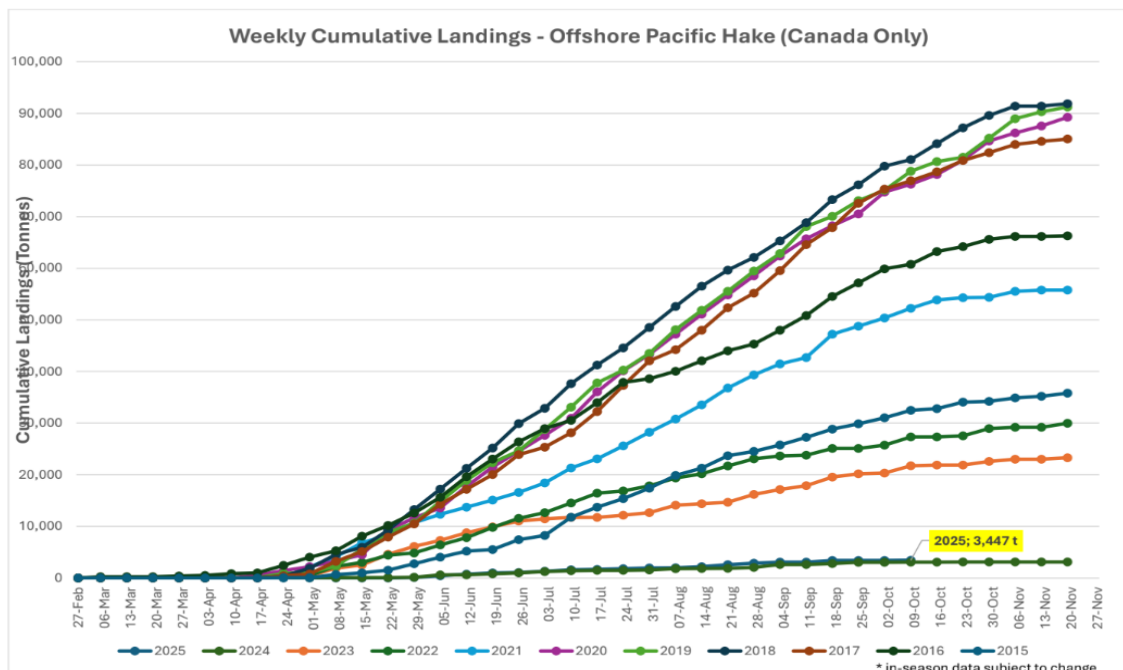


Figure E.1. Weekly cumulative landings of Pacific Hake as reported by the Canadian Advisory Panel (AP) from 2015 to 2025.

F HAKE-SPECIFIC ECOSYSTEM SUMMARY

Contributed by Mary E. Hunsicker and Andrew M. Edwards

The base model for this stock assessment includes year-specific maturity-at-age that explicitly accounts for changes in temperature, and empirical weight-at-age data that may be due to ecosystem effects. As such, the assessment is consistent with an ‘Ecosystem Approach to Fisheries Management’.

Section [F.1](#) discusses relevant ecosystem information that has been previously found to influence Pacific Hake recruitment ([Vestfals et al. 2023](#)). We focus on easily-available and updatable information, and what is presented here is not exhaustive. This approach was introduced in the 2025 stock assessment ([Johnson et al. 2025](#)), and is updated here. It should be considered as an incrementally-improving operational (updatable and expanded upon every year) hake-specific ecosystem summary. Section [F.2](#) then describes broader ecosystem and environmental conditions relevant to Pacific Hake. A risk table ([Table F.2](#)) summarizes conditions, uncertainty, and concerns related to Pacific Hake.

F.1 Ecosystem variables relevant for recruitment

We present some of the indicators that [Vestfals et al. \(2023\)](#) and subsequent analyses ([Marshall et al., pers. comm.](#)) found to correlate with higher Pacific Hake recruitment deviations ([Table F.1](#)). [Vestfals et al. \(2023\)](#) used estimated hake recruitment deviations from 1980 to 2010, based on results in the [Grandin et al. \(2020\)](#) hake assessment and ocean model variables from the University of California, Santa Cruz Regional Ocean Modeling System (ROMS) hindcast. In 2025 we conducted a preliminary update using a newer ocean modeling product (GLORYS, described below) that extends the analysis to a more recent time period (1993 to 2023), and used extended statistical methods that focus on within-sample and out-of-sample prediction for variable selection ([K. Marshall, pers. comm.](#)).

Overall, this resulted in 11 ecosystem indices described in [Table F.1](#) that have been found previously to correlate with recruitment deviations of Pacific Hake. These include three of the five indices included in the best-fitting model of [Vestfals et al. \(2023\)](#) (see their [Figure 7](#)), and three of the further nine indices found in their top 16 candidate models (their [Table 2](#)). Time series for five indices are based on values amalgamated in the *pacea* R package ([Edwards et al. 2024](#)). The other six indices were calculated from outputs of the Global Ocean Physics Reanalysis ([GLORYS12V1](#)) product, which is a global ocean eddy-resolving oceanographic model covering 1993 onwards. Details of spatial, depth, and temporal ranges are given in [Table F.1](#).

The following variables are taken from *pacea*. Pacific Herring (*Clupea pallasii*) spawning biomass index was calculated from updated results ([DFO 2025](#)) of the stock assessment, which is conducted using a statistical catch-at-age model ([Cleary et al. 2019](#)). The North Pacific Current Bifurcation Index (BI) was developed by [Malick et al. \(2017\)](#), who found that a northward-shifted bifurcation was associated with increased salmon productivity in British Columbia and Washington State waters. The index is [updated annually](#) ([Michael Malick, pers. comm.](#)), though 2025 values are not yet available. The Pacific Decadal

Table F.1. Descriptions of easily available ecosystem variables that were found to influence Pacific Hake recruitment either by Vestfals et al. (2023) or our updated analyses (unpublished). Variables are in order of life-history stage, following Table A1 of Vestfals et al. (2023); see that reference for full details. Shelf break is considered between 100 and 2,000 m isobaths, defining the longitudinal extent for several variables. The first subscripts refer to life-history stage ('pre' is preconditioning, while JA is Jan-Apr, in the egg to late larvae stage).

Abbreviation of variable	Definition	Effect on recruitment; expectation for an increase in variable
PREY _{pre,her}	Pacific Herring spawning biomass off the west coast of Vancouver Island	Increased competition with herring on summer feeding grounds leads to poorer feeding conditions and reduced adult condition; ↓ recruitment the following year
BI _{pre}	North Pacific Current Bifurcation Index	Northward-shifted bifurcation leads to increased advection of prey southwards, leading to poorer feeding conditions off British Columbia, Washington, and Oregon, and reduced adult condition; ↓ recruitment the following year
PDO _{pre}	Pacific Decadal Oscillation during preconditioning (Apr-Sep)	Indicator of basin-scale processes, negative phase linked to higher productivity and improved adult condition (Apr-Sep); ↓ recruitment the following year
NPGO _{pre}	North Pacific Gyre Oscillation during preconditioning (Apr-Sep)	Indicator of basin-scale processes, positive phase linked to higher nutrient concentrations and productivity, and improved adult condition; ↑ recruitment the following year
TEMP _{spawn}	Mean temperature during spawning (shelf break, Jan-Mar, 130-500 m depth, 31-36° N)	At higher temperatures, fish are less likely to spawn but growth rate of larvae increases; U-shaped relationship with recruitment that year
AST _{eggs}	Net along-shore transport (shelf break, Jan-Mar, 40-60 m depth, 31-36° N)	Increased northward advection away from juvenile nursery areas decreases recruitment; ↓ recruitment that year
MLD _{yolk}	Mean mixed layer depth (shelf break Jan-Apr, 31-36° N)	Larvae aggregate at base of mixed layer so mixed layer depth may limit how far they rise in the water column affecting later transport; ↓ recruitment that year
SSH _{JA.c}	Average sea-surface height off California as an indicator of basin-scale processes (from coast to 30 km offshore, Jan-Apr, 34.5-42.5° N)	Higher sea surface is indicative of higher productivity and better conditions for copepods; ↑ recruitment that year
MLD _{latelarv}	Mean mixed layer depth (shelf break Mar-Jun, 31-37° N)	Larvae aggregate at base of mixed layer so mixed layer depth may limit how far they rise in the water column affecting later transport, and possibly less competition and predation when mixed layer shallower; ↑ recruitment that year
PU _{latelarv}	Strength of poleward undercurrent (from coast to 275 m isobath, Mar-Jun, 75-275 m depth, 33.5-34.5° N)	Increased northward advection away from juvenile nursery areas decreases recruitment; ↓ recruitment that year
PRED _{age0.age1hake}	Biomass of age-1 Pacific Hake (from this stock assessment)	Age-1 hake predate on pelagic juveniles (roughly Apr-Sep); ↓ recruitment that year

Oscillation (PDO) is a monthly index which is a long-lived El Niño-like pattern of Pacific climate variability (Mantua et al. 1997). The North Pacific Gyre Oscillation (NPGO) is a climate pattern that is significantly correlated with fluctuations of salinity, nutrients, and chlorophyll-a in long-term observations in the California Current and Gulf of Alaska (Di Lorenzo et al. 2008). The Pacific Herring, PDO and NPGO variables all have an extra year of values compared to those shown in the 2025 Pacific Hake stock assessment (Johnson et al. 2025).

The estimates of age-1 Pacific Hake are taken from the current assessment model. Since the current base model did not involve updated data, the recruitment estimates are unchanged from the 2025 base model, and so the index of age-1 Pacific Hake is the same as for the 2025 assessment.

The six further indices were calculated for the 2025 stock assessment from GLORYS outputs by K. Marshall, with assistance of several colleagues in the extraction of model results, wrangling, and analysis. For 2026 we have streamlined the process of obtaining outputs from GLORYS, such that three of the GLORYS products used here can now be extracted purely using R code (soon to be available in pacea), rather than requiring use of a more complex python and R setup (thanks to Megan Feddern, Nick Tolimieri, and Eric Ward, all NOAA, for sharing their code for this). Thus, three indices are updated this year (namely $TEMP_{spawn}$, MLD_{yolk} , and $MLD_{latelarv}$), with the other three anticipated to be updatable for future assessments.

Estimates of Pacific Hake recruitment from this assessment are presented alongside the five indices calculated using outputs from pacea (Figure F.1) and the six indices calculated from GLORYS output (Figure F.2). All indices are standardised over the ranges of years shown (different for each figure), such that each index has a mean of zero and standard deviation of one. The x-axis corresponds to the year for which recruitment is expected to be influenced by the ecosystem variable represented by the index. For example, $PREY_{pre,her}$ is an index representing Pacific Herring spawning biomass off the west coast of Vancouver Island (Table F.1). Vestfals et al. (2023) found that increased herring biomass led to increased competition with hake on summer feeding grounds, leading to poorer feeding conditions and reduced adult condition. Pacific Hake recruitment would then be affected the following year. Thus, the $PREY_{pre,her}$ index for year 2020 in Figure F.1 represents the herring spawning biomass in 2019, because that is the biomass expected to influence hake recruitment in 2020. A similar shift is done for the other three variables that influence the adult preconditioning stage of hake.

The notation $PREY_{pre,her}$ represents herring being a prey item for adult hake in the preconditioning stage, although Vestfals et al. (2023) found a competition effect (not a predator-prey effect as originally hypothesised, based on conventional wisdom).

Some variables are plotted with a flipped y-axis, so that upwards in all plots consistently corresponds to conditions supposedly good for hake recruitment. For example, a lower herring biomass is expected to correspond to an increase in hake recruitment, and so the y-axis for the herring index in Figure F.1 is flipped, so that lower-than-average herring biomass is upwards on the plot. It is coloured blue to represent a negative herring biomass

anomaly. Similarly, red represents a positive herring biomass anomaly but values point downwards to indicate a negative potential effect on hake recruitment. A similar flip is done for all variables for which a negative index was found to correspond to positive hake recruitment, to make it easier to visualise potential effects. A similar approach is taken for Figure F.2.

In the analogous figures in the 2025 Pacific Hake stock assessment we showed recruitments for recent years (specifically 2023, 2024, and 2025). But in hindsight these should not be shown because they were not fit to data and so just represented median hake recruitment with no recruitment deviations. Given our current recruitment estimates are based on those from the 2025 assessment, we omit the recent years from the figures. This highlights that the stock assessment model has no information on recruitment in those years (whereas relevant ecosystem information is available) and motivates continued investigation into potential drivers of recruitment.

We have not yet conducted further analyses on these time series, but present them as an updated attempt to compile varied ecosystem information into single figures that can stimulate thinking about ecosystem effects, and that can be updated and expanded upon for future assessments. Note that the strongest driver of hake recruitment found by Vestfals et al. (2023) was eddy kinetic energy between May and September (during the female spawning preconditioning stage) from point Conception to Cape Blanco, but an up-to-date index of this was not easily available, though should be for next year (in collaboration with Wei Cheng, NOAA).

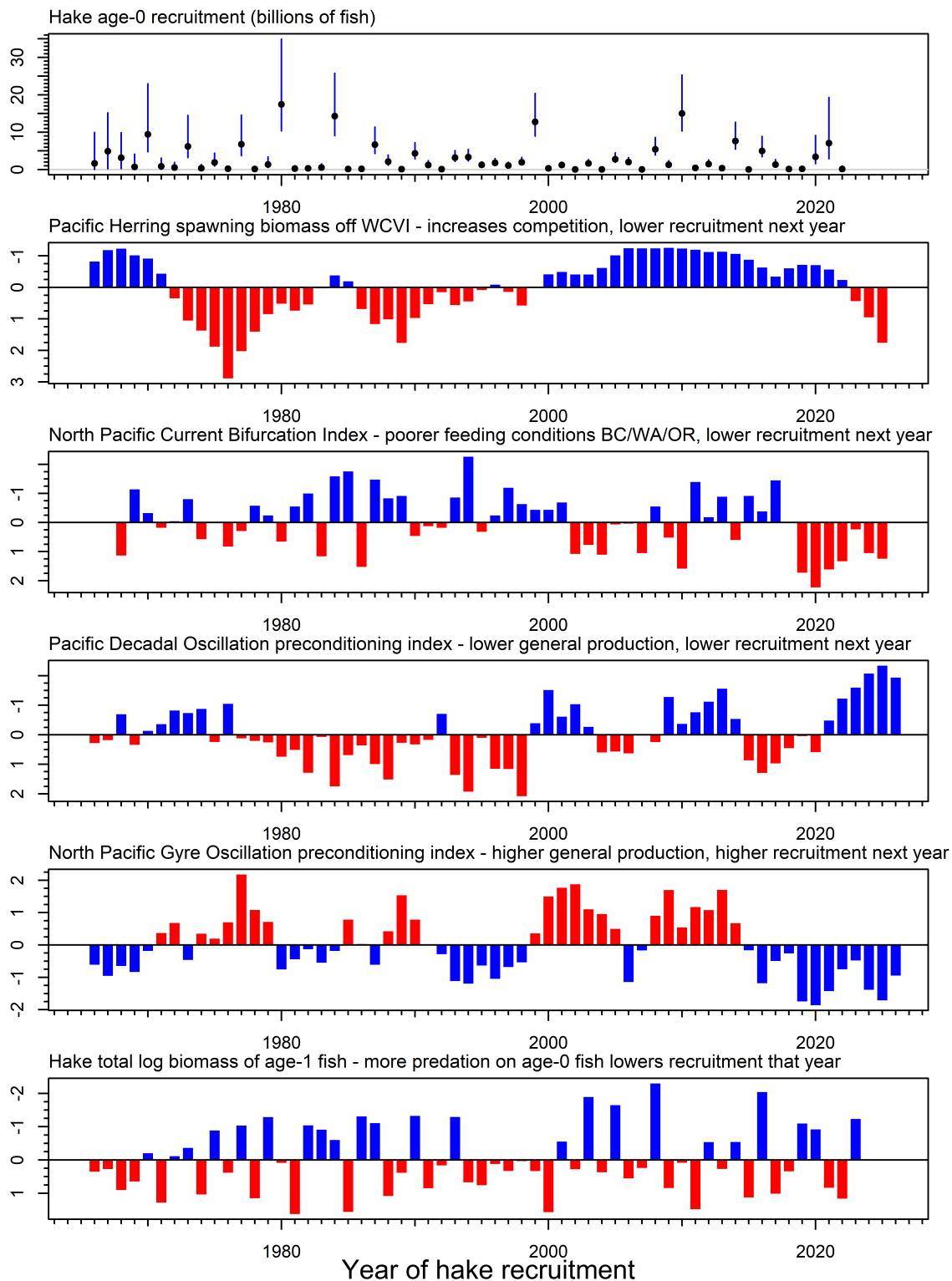


Figure F.1. Estimated hake recruitment from this assessment (top), plus five ecosystem indices based on the pacea R package. The x-axis corresponds to the expected influenced year of Pacific Hake recruitment (see text). Red (blue) bars represent a positive (negative) index, with the y-axis flipped for four indices so that upwards represents a positive expected influence on recruitment.

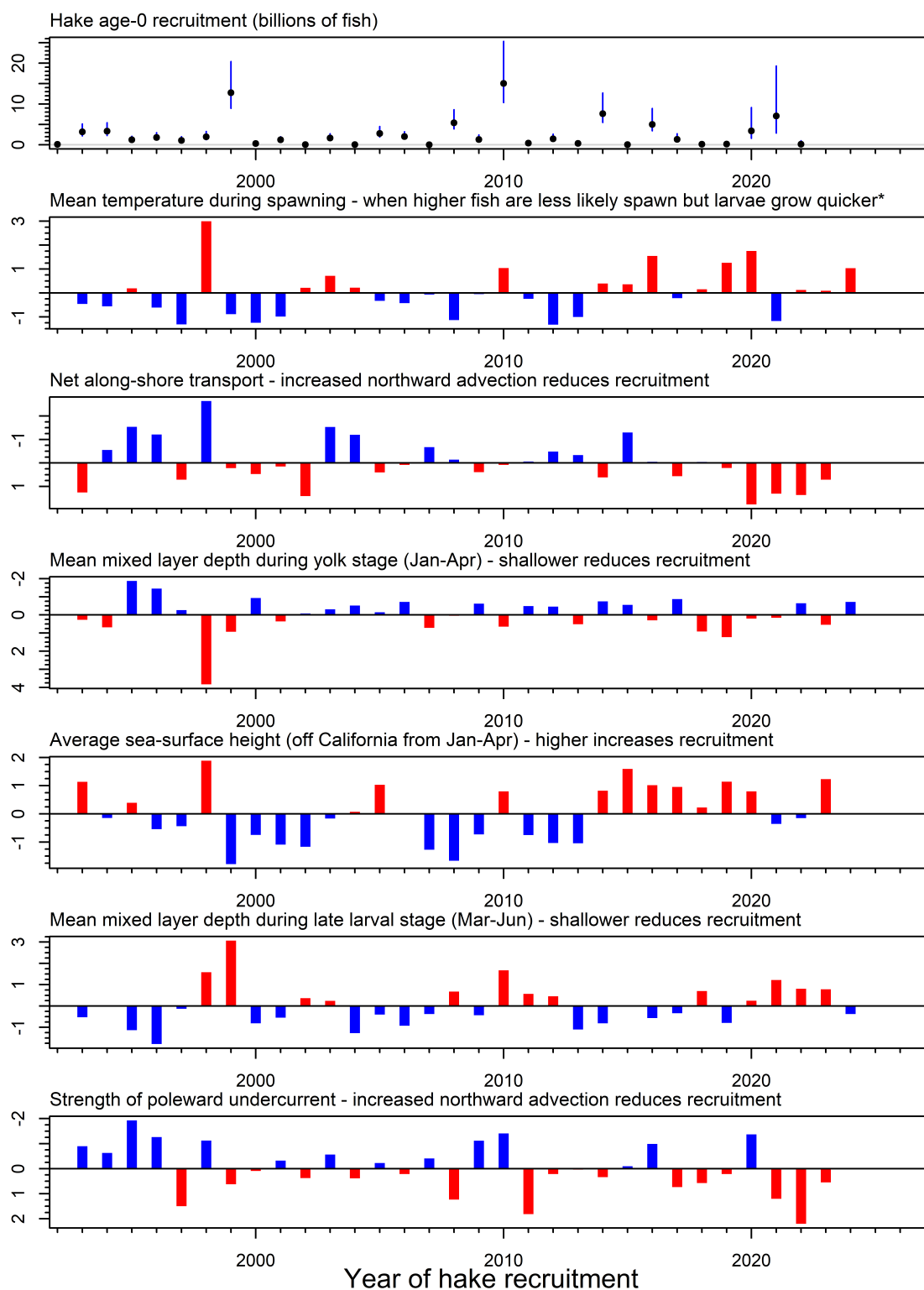


Figure F.2. Estimated Pacific Hake recruitment from the current assessment model (top), plus six ecosystem calculated from GLORYS output. Axes are defined as in Figure F.1 and in the text. *Recruitment has a U-shaped relationship with mean temperature during spawning and for simplicity we do not flip the axis.

F.2 Ecosystem and environmental conditions

To start the process of discussing the vulnerability of Pacific Hake to climate change, we evaluated recent trends in environmental drivers of hake recruitment and growth, predators, and prey, along with the climate vulnerability assessment (CVA) rank assigned to hake by McClure et al. (2023). We did not consider competitors, habitat, or non-fisheries human activities (such as offshore wind development) during this evaluation. Overall, we consider ecosystem and environmental conditions to be neutral (Level 2) for hake, with medium to high confidence, based on medium agreement among indicators and robust evidence. We use this, plus information related to this stock assessment, to fill out the ‘risk table’ in Table F.2, based on the framework outlined by the [California Current Integrated Ecosystem Assessment \(CCIEA\) team](#).

Recruitment

Climate, ocean, and ecological indicators suggest that ocean productivity off California was near to above average in 2025. The ecosystem experienced a mild La Niña in winter and spring, strong and consistent upwelling from winter through summer, and cool coastal water temperatures throughout much of the year. A strong marine heatwave caused widespread warmer-than-normal surface temperatures in offshore waters during the summer, that moved onshore in the late summer and fall. However, the anomalously warm temperatures are not reflected in the ecological indices. The biomass of lipid-rich northern copepods and indicators of krill abundance were near to above average levels in the central and northern regions of the California Current. Overall, environmental conditions in winter and spring were likely favorable for hake productivity in summer and fall. Observations of larval hake off southern California remained above average from 2021 to 2025 and juvenile hake from central California were near or above average over the same period.

In 2026, the ecosystem is expected to remain neutral with respect to El Niño conditions through summer. However, warmer-than-average temperatures are still prevalent along the coast and a marine heatwave is forecasted to continue in offshore waters until early spring. It is uncertain whether these conditions will negatively impact hake recruitment.

Hake distribution

Seasonal forecasts of hake distribution in the northern region of the California Current Ecosystem (Malick et al. 2020) were not available at the time of writing this summary, however these temperature-driven forecasts will be presented at the 2026 SRG meeting. Given what we know about the influence of temperature on hake distribution and the role of temperature as a proxy for prey availability (Malick et al. 2020; Phillips et al. 2022), we would have expected a normal extent of the northern migration of hake in the spring of 2025 as feeding conditions were favorable in the central and northern regions of the California Current.

Predators and prey

Pacific Hake are common in many predators’ diets (e.g., Bluefin Tuna, Swordfish, sharks, marine mammals, Sablefish, Arrowtooth Flounder, and adult hake). Recent trends (i.e.,

Table F.2. ‘Risk table’ for Pacific Hake, to document ecosystem and climate factors potentially affecting stock productivity and uncertainty or other concerns arising from the stock assessment (see text). Level 1 is a favourable ranking, Level 2 neutral, and Level 3 is unfavourable. CVA is the Climate Variability Assessment approach (McClure et al., 2023).

Ecosystem and environmental conditions	Assessment data inputs	Assessment model fits and structural uncertainty
• Recruitment: 2021-2025 recruitment indicators neutral to favorable • Prey: favorable (krill, juvenile hake) • Predators: unfavorable (increasing) • CVA rank: low	• Very reliable catch reporting • Generally well-sampled fishery-dependent and -independent age compositions • Informative age-2+ fishery-independent survey biomass index every other year • Includes externally estimated time-varying weight-at-age (growth) and time-varying and temperature-dependent maturity (fecundity) as data inputs • New data was limited to 2025 catch for the 2026 assessment due to the delay in availability of the 2025 survey data	• Fully Bayesian stock assessment, integrating over multiple sources of parametric uncertainty • High recruitment variability and no information on recent recruitment in assessment model from 2023 onwards, given no survey in 2024 • Key demographics estimated using priors (natural mortality and steepness) • Clear identity of large cohorts after at least age-3 • Poor model fit to the most recent (2023) survey data point • Model fits well to age compositions, though occasionally at the expense of the survey index of abundance • Uncertain how changes in the distribution of fishing relate to migration patterns and stock abundance • For 2026 the model was not refit to new data, but rather based on the 2025 assessment model, so uncertainty is inherently increased
Level 2 (medium agreement, robust evidence)	Level 2 (due to unavailability of 2025 data)	Level 2

over the last five years) in many of these populations of hake predators are stable or increasing, suggesting mild concern about increased predation. For example, the consumption of hake by Bluefin Tuna and Swordfish has been well above average in recent years, with record high levels of hake in bluefin diets in 2024. The spawning biomass of these species has increased over this period as well (Leising et al., in prep).

However, indices of prey abundance have been favorable over the past few years. Krill is a dominant diet item for juvenile and adult hake (Wippel et al. 2017; Phillips et al. 2022; Bizzarro et al. 2023). The CCIEA regional indicators show that krill has been abundant in the central CCE in recent years, and increased in waters off Oregon, Washington, and British Columbia in 2026 compared to lower abundances in those regions in 2021 and 2023.

Climate vulnerability assessment results

Hake are highly exposed with low sensitivity to climate change, with an overall rank of low (McClure et al. 2023). However, we note that the CVA work pre-dates recent studies on the relationships between hake distribution and recruitment and ocean conditions (Malick et al. 2020; Vestfals et al. 2023).

G SRG REQUESTS

G.1 Day 1

G.1.1 Request 1:

SRG request: Summarize the number of aged fish by month for each sector in the US and Canada (if possible). Compare these to the monthly catch plots.

Justification: This will help to determine if the age data are representative of the fishery catches. If the number of ages are consistent with the catch, then the ages are likely to be representative.

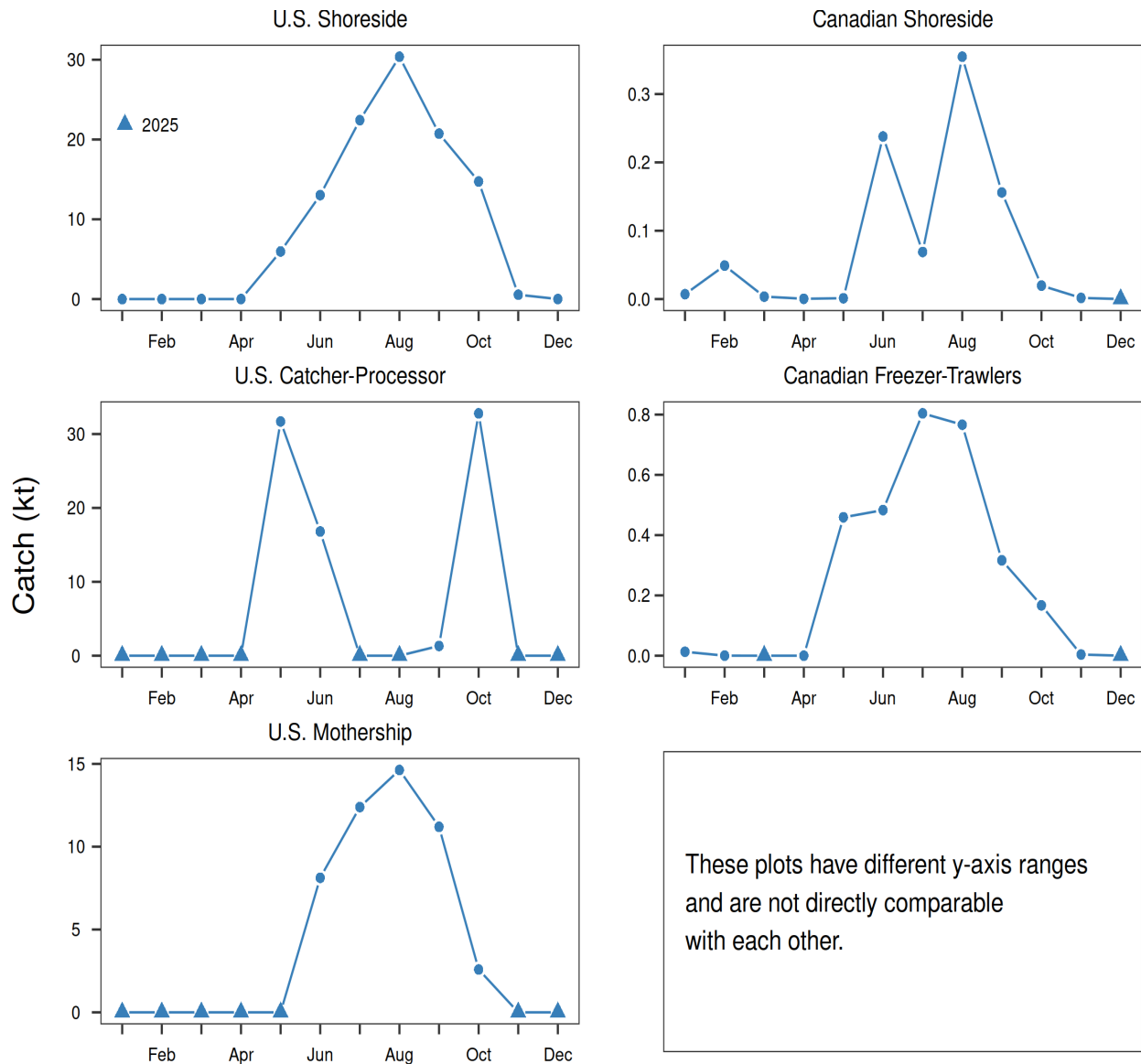


Figure G.1. Catch by month in 2025 for all sectors of the U.S. and fleets of Canada.

G.1.1.1 Request 1 Results: Sector/Fleet-based Samples by Month

Canadian Fleets

Table G.1. Canadian sampling by month for 2025 in the Freezer trawler fleet

Month	Number of ages	Number of lengths	Number of weights
May	26	52	52
Jun	24	213	213
Jul	–	60	60
Total	50	325	325

Table G.2. Canadian sampling by month for 2025 in the Shoreside fleet.

Month	Number of ages	Number of lengths	Number of weights
Jun	197	371	371
Jul	–	50	50
Aug	–	100	100
Total	197	521	521

U.S. Sectors

Table G.3. U.S. sampling by month for 2025 in the catcher-processor at-sea sector.

Month	Number of ages	Number of lengths	Number of weights
May	250	507	507
Jun	135	269	269
Jul	0	0	0
Aug	0	0	0
Sep	0	0	0
Oct	170	541	541
Total	555	1317	1317

Table G.4. U.S. sampling by month for 2025 in the mothership at-sea sector.

Month	Number of ages	Number of lengths	Number of weights
May	0	0	0
Jun	134	266	266
Jul	164	311	311
Aug	177	319	319
Sep	150	276	276
Oct	38	64	64
Total	663	1236	1236

Table G.5. U.S. sampling by month for 2025 in the shore-based sector.

Month	Number of ages	Number of lengths	Number of weights
April	2	6	6
May	118	258	167
Jun	229	480	420
Jul	450	680	560
Aug	370	680	560
Sep	290	640	490
Oct	170	508	448
Nov	0	60	40
Total	1629	3323	2708

G.1.2 Request 2

SRG request: Show the expected fishery age proportions for 2025 and 2026 fishery (this is fig 43 in 2025 assessment and slide 10 of management outcomes presentation). Possibly overlay this with 2025 fishery observations to compare the two. The JTC can decide how to present the observed 2025 fishery data.

Justification: This request will identify the correct predictions of fishery proportions-at-age for 2025 and 2026, thus providing insights into what the model expects for 2025 fisheries proportions-at-age. Comparing to the observations from 2025 will help to identify potential biases in the assessment, in particular related to the assumption of fixing recent recruitment to less than average recruitment.

G.1.2.1 Request 2 results:

Table G.6. Vulnerable projected age compositions for 2026.

Age	Cohort	By number	By weight
15	2011	0%	0%
14	2012	0%	1%
13	2013	0%	0%
12	2014	3%	5%
11	2015	0%	0%
10	2016	4%	6%
9	2017	2%	3%
8	2018	0%	1%
7	2019	1%	2%
6	2020	19%	22%
5	2021	52%	46%
4	2022	2%	2%
3	2023	6%	4%
2	2024	5%	3%
1	2025	3%	0%

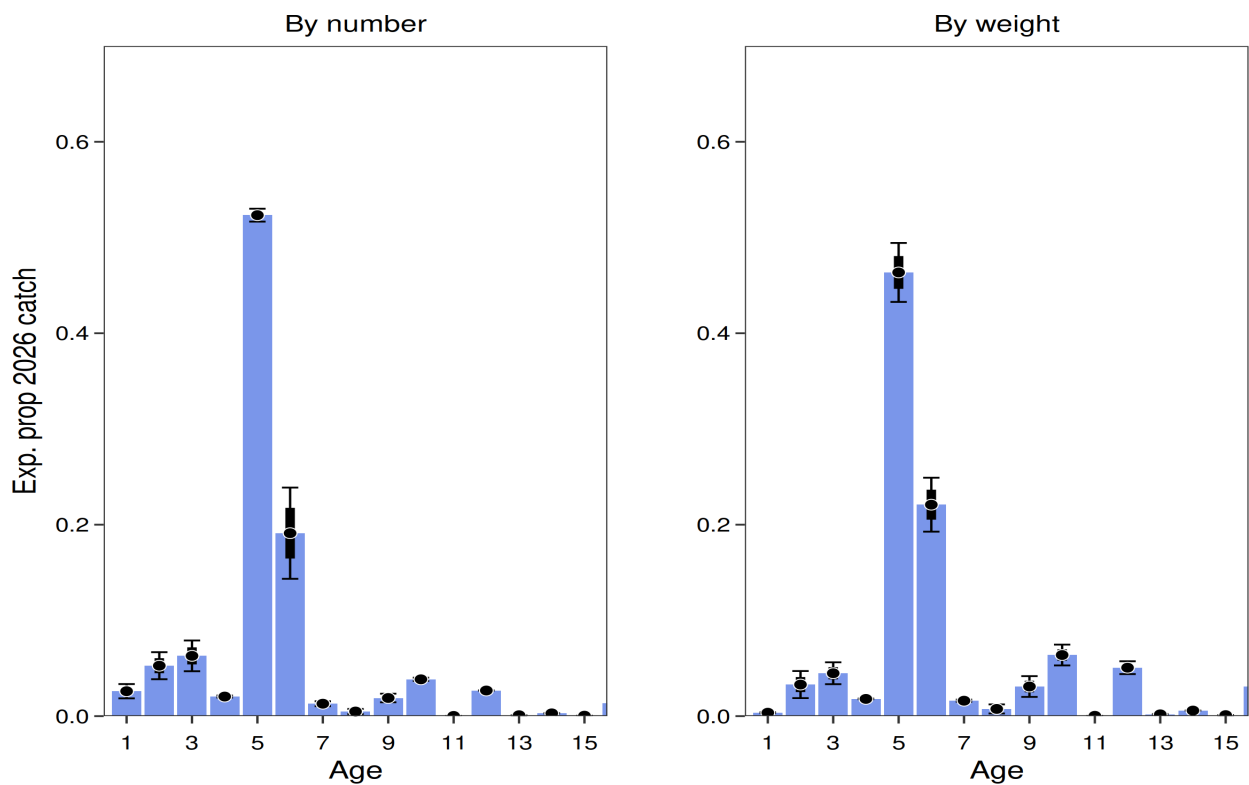


Figure G.2. Vulnerable projected age compositions for 2026. The left panel shows the proportions by number; the right panel shows the proportions by weight. Selectivity for 2026 was not available in the model output so 2025 was used instead.

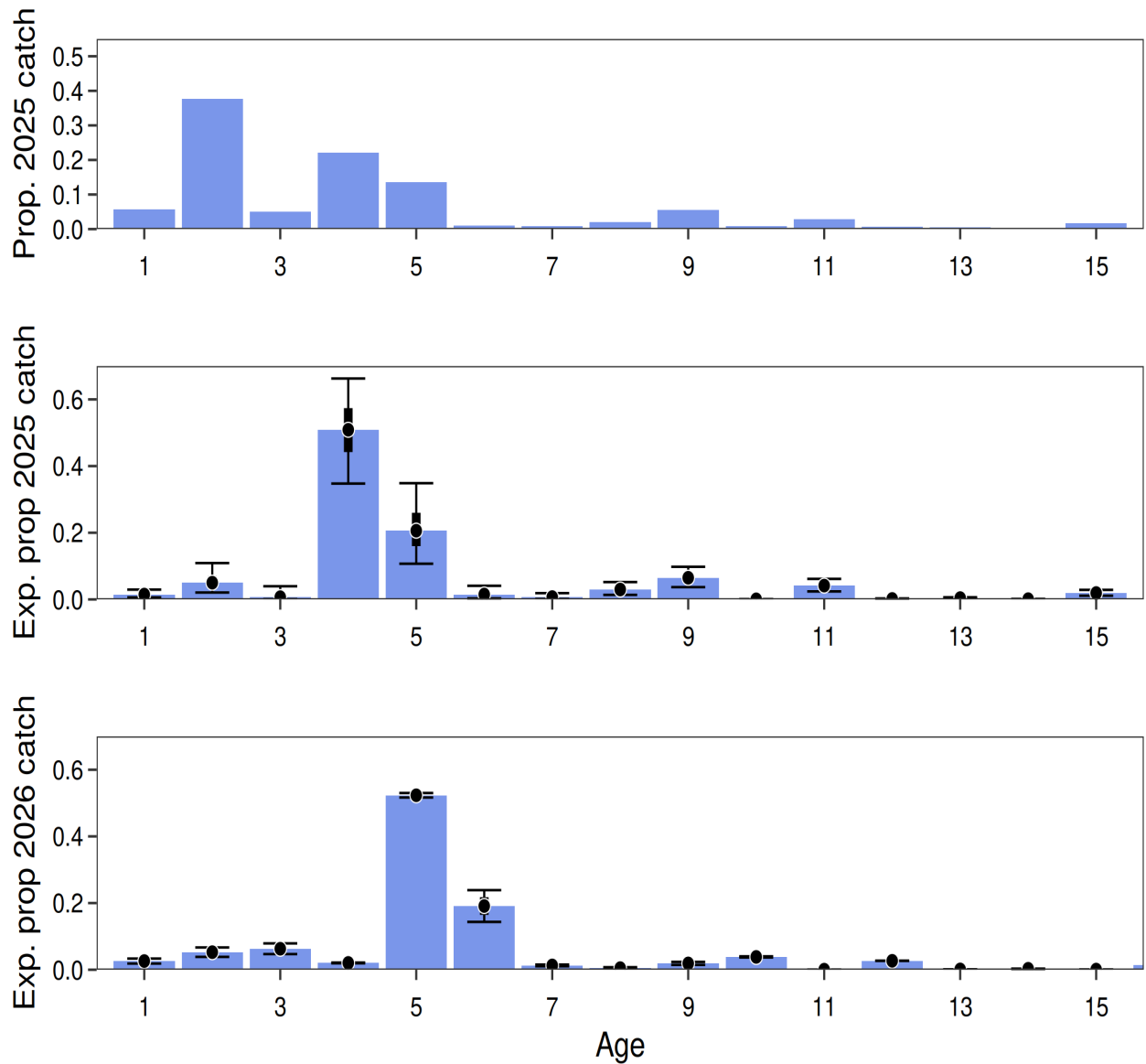


Figure G.3. Proportions of 2025 catch-at-age, expected proportions of catch-at-age from the 2025 assessment model, and expected proportions of catch-at-age from the 2026 assessment model.

G.1.3 Request 3

SRG request: Present the table of spawning biomass, age 2+ biomass and other quantities as in Table 23 from the 2025 assessment, and include this updated version in the 2026 assessment document.

Justification: These quantities will help provide insight into the possible predicted size of the 2025 acoustic survey biomass.

G.1.3.1 Request 3 results:

Table G.7. Time series of median posterior population estimates from the base model. Relative spawning biomass is spawning biomass relative to the unfished equilibrium (B_0). Total biomass includes females and males of ages 0 and above. Age-2+ biomass includes females and males ages 2 and above. Exploitation fraction is total catch divided by total age-2+ biomass. Relative fishing intensity is $(1 - \text{SPR}) / (1 - \text{SPR}_{40\%})$ such that values below 100% represent fishing below $F_{40\%}$. In the last row, dashes (–) indicate quantities requiring 2026 catch which has not taken place yet.

Year	Female spawning biomass (kt)	Relative spawning biomass (%)	Total biomass (kt)	Age-2+ biomass (kt)	Age-0 recruits (millions)	Relative fishing intensity (%)	Exploitation fraction (%)
1966	956	52.7	3,716	3,323	1,697	48.0	6.4
1967	968	53.5	4,123	3,528	4,887	65.0	9.7
1968	978	54.1	4,669	3,940	3,182	46.8	5.4
1969	1,119	62.2	5,311	4,884	726	57.1	6.5
1970	1,284	71.0	5,704	5,292	9,562	62.7	7.7
1971	1,313	72.8	6,033	4,881	865	46.6	5.5
1972	1,495	82.7	6,701	6,514	551	35.1	3.0
1973	1,779	98.1	6,800	6,511	6,277	38.5	4.1
1974	1,722	95.0	6,582	5,841	355	45.1	5.9
1975	1,486	82.0	5,651	5,586	1,908	55.0	6.4
1976	2,146	118.1	7,586	7,352	209	42.5	5.2
1977	2,074	114.0	7,250	6,991	6,821	28.3	3.0
1978	1,716	94.3	6,198	5,440	133	27.6	3.0
1979	1,848	101.7	6,676	6,629	1,399	31.7	3.1
1980	1,747	96.3	5,953	5,399	17,645	25.2	2.4
1981	1,673	92.4	6,438	4,782	280	36.7	4.1
1982	1,812	99.6	6,802	6,777	319	31.5	2.1
1983	2,324	127.9	6,764	6,734	552	26.4	2.2
1984	2,344	128.6	6,496	6,093	14,342	29.2	2.9
1985	2,349	128.7	7,334	5,836	134	23.1	2.4
1986	2,359	129.5	7,684	7,659	186	34.0	3.4
1987	2,418	132.8	6,763	6,574	6,718	38.3	4.4
1988	2,390	131.0	6,658	5,953	2,130	40.6	5.1
1989	2,076	113.7	5,998	5,829	115	48.6	6.3
1990	2,051	112.4	5,594	5,464	4,383	40.6	5.8
1991	1,861	102.1	5,021	4,581	1,245	58.9	8.3
1992	1,694	92.9	4,699	4,544	126	57.4	7.8
1993	1,326	72.5	3,527	3,447	3,186	46.4	6.9
1994	1,280	70.0	3,486	3,108	3,378	59.3	13.8
1995	1,137	62.2	3,458	3,048	1,249	51.4	9.6
1996	1,101	60.3	3,321	3,186	1,821	63.5	11.5
1997	1,096	60.0	3,076	2,910	1,084	67.8	13.3
1998	912	50.0	2,514	2,363	1,961	81.3	15.7
1999	784	43.1	2,442	2,016	12,796	91.7	18.0
2000	909	49.8	3,728	2,333	311	65.1	11.3
2001	1,339	73.6	5,032	4,949	1,241	66.9	5.4
2002	1,909	104.9	5,246	5,106	32	47.6	4.1
2003	1,785	98.1	4,324	4,271	1,668	42.6	5.5
2004	1,453	79.8	3,466	3,296	55	68.8	11.6
2005	1,152	63.4	2,813	2,749	2,777	66.6	14.8
2006	915	50.4	2,384	2,106	2,016	85.0	19.0
2007	721	39.7	1,996	1,870	24	86.0	17.6
2008	647	35.7	2,057	1,948	5,437	96.0	18.7
2009	603	33.3	1,880	1,447	1,318	80.1	14.0
2010	722	39.9	2,525	2,100	15,143	89.8	12.7
2011	716	39.7	3,259	2,021	416	88.9	16.5
2012	886	49.1	4,337	4,262	1,487	73.7	5.9
2013	1,622	89.9	4,889	4,723	352	69.1	7.3
2014	1,883	104.5	5,002	4,764	7,685	64.7	7.5
2015	1,377	76.1	3,899	3,273	31	47.9	7.0

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Year	Female spawning biomass (kt)	Relative spawning biomass (%)	Total biomass (kt)	Age-2+ biomass (kt)	Age-0 recruits (millions)	Relative fishing intensity (%)	Exploitation fraction (%)
2016	1,116	61.7	3,844	3,729	5,048	79.1	10.4
2017	1,510	83.5	4,177	3,650	1,342	83.9	14.0
2018	1,582	87.7	4,360	4,190	158	77.3	11.2
2019	1,282	71.0	3,432	3,410	191	86.3	13.6
2020	1,181	65.5	2,826	2,718	3,473	68.6	15.6
2021	947	52.3	2,619	2,126	7,182	70.5	17.3
2022	932	51.3	2,995	2,328	133	73.3	14.9
2023	1,128	61.9	3,211	3,155	886	65.5	9.1
2024	1,203	66.0	3,237	3,108	890	66.0	5.9
2025	1,240	67.4	2,955	2,825	892	63.2	9.3
2026	1,066	57.5	2,596	2,466	877	–	–

G.1.4 Request 4

SRG request: Create and present the plot showing the amount of each cohort in the catch over time (age on the x-axis). This plot has been created in the past. Please include the 1999, 2010, 2014, 2016, 2017, 2020, and 2021 cohorts.

Justification: This plot shows the rate of removals of a cohort over time and how it compares across cohorts, which is helpful to understand the importance of a cohort to the fishery.

G.1.4.1 Request 4 results:

Estimated total catch from each cohort

- 1999 cohort (ages 1–26): 2,179,715 t
- 2010 cohort (ages 1–15): 2,596,031 t
- 2014 cohort (ages 1–11): 1,773,735 t
- 2016 cohort (ages 1–9): 1,099,470 t
- 2017 cohort (ages 1–8): 293,786 t
- 2020 cohort (ages 1–5): 335,625 t
- 2021 cohort (ages 1–4): 312,198 t

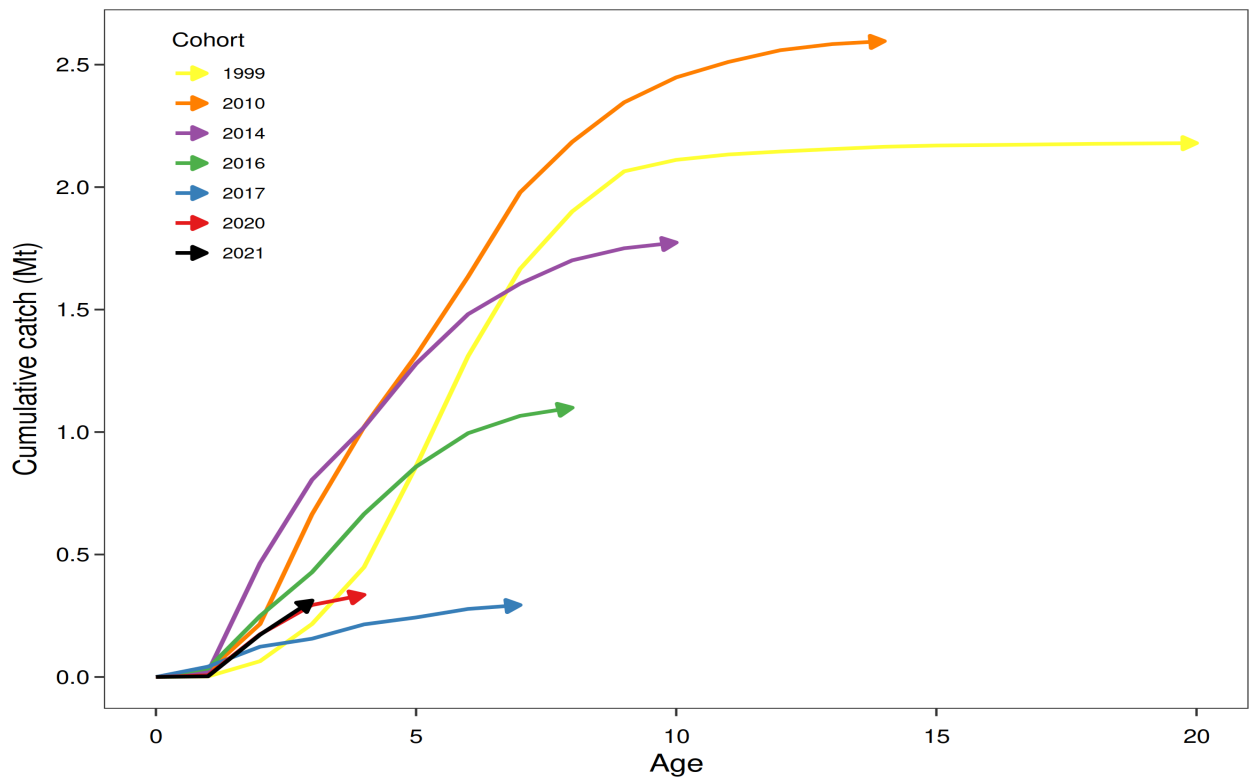


Figure G.4. Cumulative catch of key cohorts as requested by the SRG.

G.1.5 Request 5

SRG request: Calculate and present probability tables (slide 15 of management outcomes) for each sensitivity run (at least the no additional standard error on the survey and the stochastic recruitment runs).

Justification: This will provide an insight into the management outcomes for alternative model runs and identify the importance (through the size of the effect) to management outcomes.

G.1.5.1 Request 5 results:

Table G.8. Probabilities based on 2026 catch (Base Model).

	Catch (t) in 2026	$B_{2027} < B_{2026}$	$B_{2027} < B_{40\%}$	$B_{2027} < B_{25\%}$	$B_{2027} < B_{10\%}$	2026 Fishing intensity > 100%	2027 Default HR catch < 2026 catch
a:	0	0.85	0.21	0.02	0.00	0.00	0.00
b:	150,000	1.00	0.28	0.05	0.00	0.01	0.01
c:	200,000	1.00	0.31	0.06	0.00	0.02	0.03
d:	250,000	1.00	0.33	0.07	0.00	0.05	0.08
e:	300,000	1.00	0.36	0.09	0.00	0.10	0.15
f:	350,000	1.00	0.39	0.11	0.00	0.15	0.23
g:	400,000	1.00	0.41	0.13	0.01	0.21	0.33
h:	450,000	1.00	0.44	0.15	0.01	0.28	0.42
i:	500,000	1.00	0.47	0.17	0.01	0.35	0.51
j:	559,334	1.00	0.50	0.20	0.01	0.42	0.60

Table G.9. Probabilities based on 2026 catch (Stochastic forecast recruitment).

	Catch (t) in 2026	$B_{2027} < B_{2026}$	$B_{2027} < B_{40\%}$	$B_{2027} < B_{25\%}$	$B_{2027} < B_{10\%}$	2026 Fishing intensity > 100%	2027 Default HR catch < 2026 catch
a:	0	0.42	0.03	0.00	0.00	0.00	0.00
b:	150,000	0.58	0.05	0.01	0.00	0.00	0.00
c:	200,000	0.62	0.06	0.01	0.00	0.01	0.01
d:	250,000	0.66	0.07	0.01	0.00	0.02	0.02
e:	300,000	0.69	0.07	0.01	0.00	0.03	0.05
f:	350,000	0.72	0.08	0.02	0.00	0.05	0.08
g:	400,000	0.74	0.09	0.02	0.00	0.07	0.13
h:	450,000	0.76	0.10	0.02	0.00	0.10	0.17
i:	500,000	0.78	0.10	0.03	0.00	0.14	0.23
j:	793,411	0.85	0.16	0.06	0.00	0.39	0.53

Table G.10. Probabilities based on 2026 catch (Stochastic forecast recruitment and Fit Survey).

	Catch (t) in 2026	$B_{2027} < B_{2026}$	$B_{2027} < B_{40\%}$	$B_{2027} < B_{25\%}$	$B_{2027} < B_{10\%}$	2026 Fishing intensity > 100%	2027 Default HR catch < 2026 catch
a:	0	0.34	0.06	0.00	0.00	0.00	0.00
b:	150,000	0.52	0.09	0.00	0.00	0.00	0.00
c:	200,000	0.56	0.10	0.01	0.00	0.00	0.00
d:	250,000	0.61	0.11	0.01	0.00	0.00	0.01
e:	300,000	0.64	0.12	0.01	0.00	0.02	0.03
f:	350,000	0.67	0.13	0.02	0.00	0.04	0.07
g:	400,000	0.70	0.15	0.02	0.00	0.08	0.13
h:	450,000	0.72	0.16	0.03	0.00	0.13	0.20
i:	500,000	0.74	0.17	0.03	0.00	0.19	0.28
j:	702,572	0.81	0.23	0.07	0.00	0.45	0.55

Table G.11. Probabilities based on 2026 catch (Fit Survey).

	Catch (t) in 2026	$B_{2027} < B_{2026}$	$B_{2027} < B_{40\%}$	$B_{2027} < B_{25\%}$	$B_{2027} < B_{10\%}$	2026 Fishing intensity > 100%	2027 Default HR catch < 2026 catch
a:	0	0.80	0.34	0.01	0.00	0.00	0.00
b:	150,000	1.00	0.49	0.02	0.00	0.00	0.00
c:	200,000	1.00	0.54	0.03	0.00	0.00	0.00
d:	250,000	1.00	0.60	0.05	0.00	0.02	0.04
e:	300,000	1.00	0.65	0.07	0.00	0.07	0.12
f:	350,000	1.00	0.69	0.09	0.00	0.16	0.25
g:	400,000	1.00	0.74	0.12	0.00	0.28	0.41
h:	450,000	1.00	0.78	0.16	0.00	0.41	0.56
i:	500,000	1.00	0.81	0.20	0.00	0.54	0.69
j:	480,204	1.00	0.80	0.18	0.00	0.49	0.64

G.1.6 Request 6

SRG request: Present the fish age throughout the 2025 season for each sector. Presenting age would remove confounding growth. Monthly would be preferred, but could be spring/summer/fall, or simply early and late, if necessary.

Justification: This will address a question from the public and inform the SRG on differences in catch throughout the season. It will also help to identify when the age-1 fish became prominent and may help to understand why certain percentages are in the data but may not be in the prediction or in what stakeholders perceived for the season.

G.1.6.1 Request 6 results:

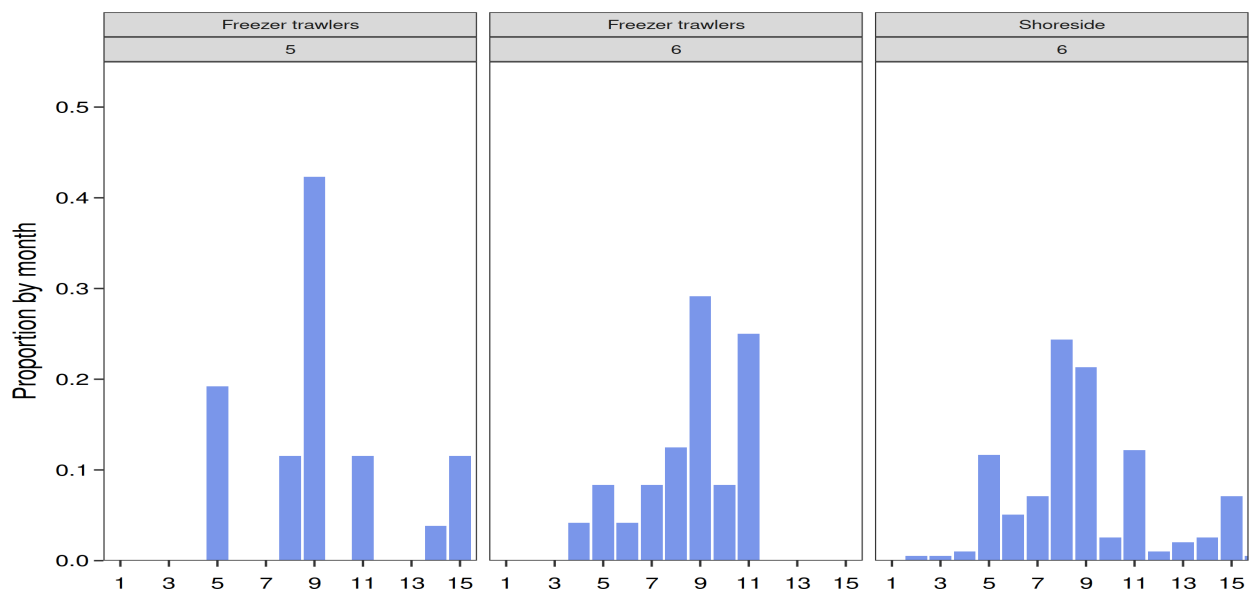


Figure G.5. Canadian proportion-at-age by month and fleet for 2025.

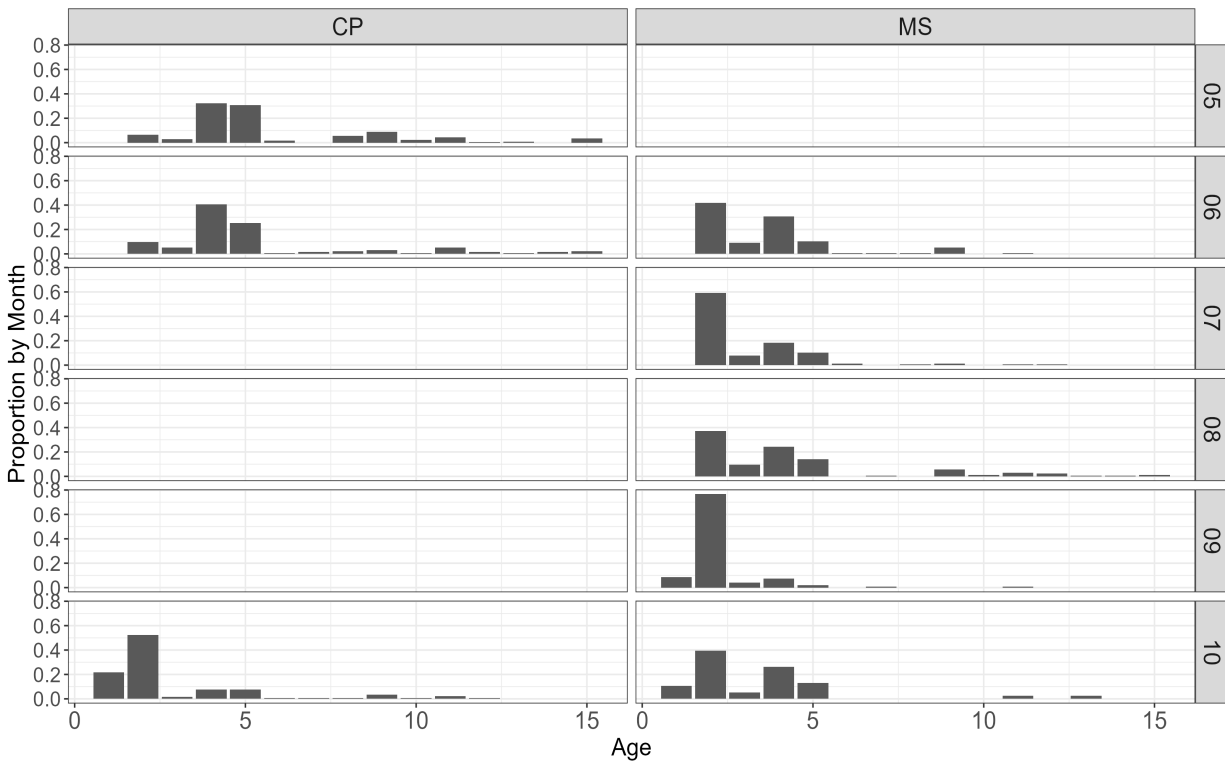


Figure G.6. U.S. at-sea proportion-at-age by month and fleet for 2025.

G.1.7 Request 7

SRG request: Extend the assessment model to incorporate data through 2025 and fit to fishery age compositions, acoustic survey biomass index, and acoustic survey age compositions. If this is not possible, or not preferable, please provide an explanation.

Justification: This will provide insight into the likely outcomes from an assessment incorporating all of the 2025 data.

G.1.7.1 Request 7 results:

The JTC supplied the following reasons for not extending the assessment model to incorporate data through 2025.

A catch-only projection is appropriate for the 2026 stock assessment for several reasons that complement the Introduction of the draft assessment document.

- The 2025 IWCPs acoustic survey biomass and age-composition estimates were finalized only days before the assessment deadline (i.e., shortly before the SRG meeting) due to a 7-week delay caused by an external time constraint (U.S. federal government shutdown). Consequently, these data have not had the necessary review by the JTC. The 2025 IWCPs survey had meaningful survey design changes. Incorporating these data without proper review by the SRG may result in assumptions that result in biased results.

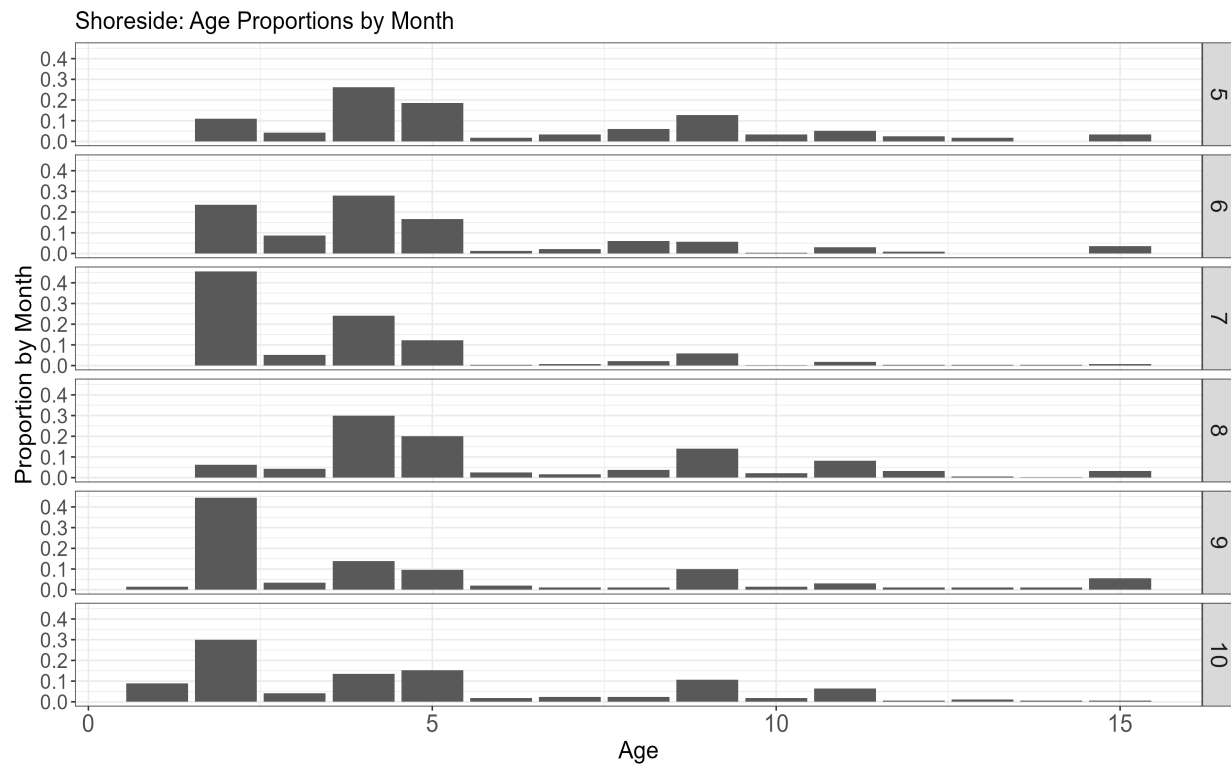


Figure G.7. U.S. shoreside proportion-at-age by month and fleet for 2025.

- Data from both the fishery and acoustic survey are complementary to provide a more informative representation of the population. Including the 2025 fishery age compositions without including the 2025 survey biomass estimate (and age compositions) risks introducing biases. In recent years, a tension has been observed between fishery age composition data (suggesting high recruitment from shifts to large proportions of young fish) and the acoustic survey biomass index (suggesting decreasing overall biomass).
- The next full stock assessment should include the full complement of properly scrutinized data informing both cohort strengths and absolute biomass.

Therefore, it is not appropriate to run new base case models for management advice during the SRG that include 2025 age composition and 2025 IWCPs biomass estimates for several reasons:

- There is insufficient time to properly review changes in new survey and trawl net designs (including selectivity and other nuances specific to the 2025 IWCPs) and then also evaluate how results from that review should be incorporated into the stock assessment.
- There is insufficient time for proper implementation of review suggestions that often require comparing alternative approaches and identifying robust options.

- There is insufficient time to properly process all of the survey-dependent data products used in a full stock assessment (including maturity-at-age, weight-at-age, and appropriately catch-weighted fishery age compositions).
- There is insufficient time to conduct the suite of sensitivities and all other diagnostic model evaluations required for a full stock assessment.

G.2 Day 3

G.2.1 Request 1:

SRG request: Present two new time series tables of spawning biomass, age 2+ biomass and other quantities as in Table 23 from the 2025 assessment. The first should be base case with stochastic recruitment and the second table is base model with no additional variance on the survey.

Justification: No official justification given.

G.2.1.1 Request Results:

Table G.12. Time series of median posterior population estimates from the model with stochastic recruitment in the projection year.

Year	Female spawning biomass (kt)	Relative spawning biomass (%)	Total biomass (kt)	Age-2+ biomass (kt)	Age-0 recruits (millions)	Relative fishing intensity (%)	Exploitation fraction (%)
1966	940	53.4	3,286	2,710	1,656	48.8	6.5
1967	956	54.3	3,980	3,340	4,790	65.6	9.7
1968	965	54.9	4,797	3,686	3,125	47.4	5.5
1969	1,101	62.8	5,752	5,572	696	58.0	6.6
1970	1,267	72.2	6,028	5,588	9,382	63.4	7.8
1971	1,301	74.3	6,201	4,982	833	47.0	5.5
1972	1,479	84.6	6,830	6,660	541	35.5	3.0
1973	1,757	100.3	6,901	6,559	6,183	39.0	4.1
1974	1,701	96.9	6,757	5,778	344	45.6	6.0
1975	1,468	83.8	5,969	5,908	1,890	55.8	6.5
1976	2,116	120.6	8,103	7,800	200	43.2	5.3
1977	2,041	116.1	7,787	7,493	6,744	28.7	3.0
1978	1,690	96.1	6,685	5,801	132	28.0	3.0
1979	1,824	103.6	7,234	7,164	1,388	32.3	3.1
1980	1,729	98.0	6,531	5,843	17,419	25.6	2.4
1981	1,658	93.8	7,222	5,173	272	37.3	4.1
1982	1,794	101.7	7,793	7,743	318	32.0	2.1
1983	2,302	130.7	7,781	7,727	537	26.7	2.2
1984	2,323	131.7	7,461	7,001	14,169	29.4	2.9
1985	2,328	132.0	8,349	6,651	130	23.4	2.4
1986	2,344	132.7	8,651	8,637	180	34.3	3.5
1987	2,409	135.8	7,534	7,328	6,653	38.6	4.4
1988	2,381	133.9	7,354	6,549	2,121	40.8	5.1
1989	2,066	116.3	6,625	6,369	111	49.0	6.3
1990	2,039	115.0	6,224	6,091	4,315	40.9	5.8
1991	1,852	104.5	5,615	5,077	1,237	59.6	8.4
1992	1,685	95.1	5,289	5,088	124	57.9	7.9
1993	1,319	74.4	4,018	3,918	3,156	46.9	6.9
1994	1,273	71.9	3,978	3,533	3,344	59.9	13.9

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Year	Female spawning biomass (kt)	Relative spawning biomass (%)	Total biomass (kt)	Age-2+ biomass (kt)	Age-0 recruits (millions)	Relative fishing intensity (%)	Exploitation fraction (%)
1995	1,131	63.9	3,947	3,532	1,250	51.7	9.7
1996	1,095	61.8	3,745	3,576	1,809	63.7	11.6
1997	1,089	61.4	3,476	3,249	1,083	68.2	13.3
1998	908	51.2	2,882	2,705	1,938	81.6	15.8
1999	781	44.1	2,843	2,357	12,697	92.1	18.1
2000	906	51.1	4,376	2,679	309	65.5	11.4
2001	1,332	75.4	5,911	5,825	1,228	67.3	5.4
2002	1,901	107.5	6,120	5,949	32	48.0	4.1
2003	1,778	100.5	5,001	4,938	1,659	42.8	5.5
2004	1,447	81.7	3,993	3,779	56	69.2	11.7
2005	1,148	64.8	3,281	3,179	2,764	67.1	14.8
2006	912	51.6	2,873	2,479	1,996	86.0	19.1
2007	717	40.6	2,527	2,344	24	86.4	17.7
2008	644	36.5	2,704	2,563	5,400	96.2	18.8
2009	601	34.1	2,540	1,979	1,298	80.5	14.0
2010	720	40.8	3,377	2,807	15,067	90.1	12.7
2011	715	40.5	4,295	2,774	415	89.2	16.6
2012	884	50.3	5,599	5,494	1,475	74.0	5.9
2013	1,618	92.1	6,213	6,019	350	69.5	7.3
2014	1,881	107.0	6,299	6,022	7,659	64.8	7.5
2015	1,376	78.2	4,854	4,079	30	48.0	7.0
2016	1,116	63.5	4,747	4,600	5,056	79.2	10.4
2017	1,512	86.0	5,195	4,508	1,342	84.3	13.9
2018	1,589	90.6	5,504	5,257	166	77.3	11.2
2019	1,287	73.3	4,395	4,375	188	85.9	13.6
2020	1,186	67.7	3,669	3,546	3,507	68.6	15.6
2021	952	54.2	3,484	2,846	7,411	70.3	17.2
2022	943	53.6	4,147	2,939	180	73.1	14.7
2023	1,153	65.1	5,214	4,676	5,024	65.0	8.8
2024	1,264	71.3	6,434	4,846	898	58.8	5.7
2025	1,615	90.1	7,796	7,590	907	54.9	6.3
2026	1,781	99.4	7,891	7,723	916	–	–

Table G.13. Time series of median posterior population estimates from the model with no additional variance on the survey.

Year	Female spawning biomass (kt)	Relative spawning biomass (%)	Total biomass (kt)	Age-2+ biomass (kt)	Age-0 recruits (millions)	Relative fishing intensity (%)	Exploitation fraction (%)
1966	1,055	56.1	2,642	2,387	2,004	43.6	5.8
1967	1,072	57.4	2,799	2,485	5,601	59.5	8.6
1968	1,099	58.9	3,027	2,534	3,818	41.7	4.8
1969	1,260	67.8	3,449	3,137	809	51.4	5.7
1970	1,454	78.4	3,785	3,481	11,381	56.3	6.7
1971	1,507	80.8	4,067	3,225	1,002	40.8	4.8
1972	1,719	91.9	4,640	4,541	656	30.3	2.6
1973	2,039	109.1	4,771	4,578	7,400	33.5	3.6
1974	1,973	105.5	4,644	4,087	418	39.3	5.2
1975	1,709	91.3	4,026	3,975	2,242	48.5	5.6
1976	2,476	131.6	5,457	5,279	244	37.0	4.5
1977	2,385	126.7	5,282	5,083	7,968	24.2	2.6
1978	1,961	104.3	4,637	3,984	159	23.6	2.6
1979	2,107	112.0	5,170	5,125	1,630	27.5	2.7
1980	1,984	105.6	4,756	4,275	20,234	21.7	2.1
1981	1,895	100.3	5,317	3,881	326	32.1	3.6
1982	2,041	108.4	5,783	5,758	371	27.4	1.9

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Year	Female spawning biomass (kt)	Relative spawning biomass (%)	Total biomass (kt)	Age-2+ biomass (kt)	Age-0 recruits (millions)	Relative fishing intensity (%)	Exploitation fraction (%)
1983	2,601	138.3	5,871	5,844	638	23.0	1.9
1984	2,606	138.5	5,707	5,384	16,173	25.6	2.6
1985	2,598	137.9	6,460	5,185	154	20.2	2.1
1986	2,601	137.9	6,776	6,763	215	30.3	3.1
1987	2,652	140.8	5,991	5,839	7,571	34.4	4.0
1988	2,610	138.4	5,926	5,318	2,487	36.5	4.7
1989	2,263	119.9	5,371	5,199	134	44.0	5.7
1990	2,242	118.6	5,065	4,960	4,825	36.7	5.3
1991	2,034	107.6	4,584	4,187	1,364	53.3	7.6
1992	1,850	97.8	4,303	4,186	160	52.1	7.2
1993	1,447	76.4	3,245	3,159	4,027	42.2	6.3
1994	1,392	73.6	3,244	2,859	3,970	54.7	12.7
1995	1,261	66.7	3,277	2,913	1,574	46.8	8.6
1996	1,250	66.2	3,191	3,035	2,001	57.5	10.1
1997	1,265	67.1	3,010	2,840	1,008	61.0	11.4
1998	1,068	56.5	2,490	2,378	1,972	72.6	13.5
1999	918	48.7	2,375	1,993	14,004	83.9	15.6
2000	1,045	55.4	3,527	2,280	393	58.1	10.0
2001	1,485	78.4	4,707	4,635	1,387	61.5	4.9
2002	2,078	109.8	4,928	4,789	38	43.1	3.8
2003	1,928	101.8	4,103	4,053	2,001	37.9	5.1
2004	1,562	82.5	3,331	3,159	70	63.3	10.8
2005	1,246	65.9	2,742	2,676	3,345	61.6	13.6
2006	1,003	53.0	2,357	2,085	2,217	79.2	17.3
2007	808	42.8	1,996	1,870	27	79.0	15.6
2008	742	39.3	2,059	1,971	5,450	88.4	16.4
2009	703	37.2	1,836	1,488	1,377	72.3	12.0
2010	810	42.9	2,333	1,968	16,201	82.4	11.6
2011	783	41.4	2,911	1,868	450	84.3	15.4
2012	950	50.3	3,806	3,744	1,585	68.9	5.5
2013	1,709	90.3	4,286	4,150	342	63.1	6.9
2014	1,973	104.3	4,385	4,181	7,949	61.1	7.2
2015	1,431	75.6	3,419	2,873	35	45.2	6.7
2016	1,151	60.8	3,397	3,290	5,368	76.7	10.1
2017	1,540	81.3	3,714	3,217	1,384	81.2	13.7
2018	1,618	85.3	3,918	3,771	154	74.7	11.0
2019	1,305	68.9	3,100	3,085	117	83.5	13.3
2020	1,197	63.1	2,528	2,457	2,653	66.4	15.4
2021	943	49.8	2,255	1,887	6,270	70.3	17.3
2022	862	45.5	2,446	1,942	134	76.9	16.7
2023	961	50.7	2,523	2,484	969	70.1	10.6
2024	993	52.3	2,479	2,377	972	67.3	7.2
2025	1,012	53.1	2,252	2,152	973	68.6	11.3
2026	864	45.2	1,948	1,849	951	–	–